

Sidewalk Sidekick: Report 3

ME/MSE 4723

Professor Stephen Sprigle

Georgia Institute of Technology

Sadhana Kumar, Sahas Suri, Pranav Nair, Vincent Griffio, Keerthi Padmanabhan, Grant Espy

Table of Contents

Nomenclature and Glossary.....	5
Executive Summary.....	6
Introduction and Background	7
Existing Products, Prior Art, and Relevant Literature	7
Stakeholders and Market Research.....	8
Customer Requirements and Engineering Design Specifications.....	8
Criteria Table	8
Key Functionality	9
Engineering Specifications	9
Codes and Standards.....	10
Design Evolution	10
Concept Generation	10
Concept Selection	10
Concept Iteration.....	11
Mechanical Architecture.....	12
Electrical Architecture	12
Software Architecture	14
Human-Robot Interface	15
Anthropometric and Ergonomic Considerations.....	15
Tactile and Visual Interaction	15
Concepts	15
User-Interface Testing and Validation	16
Feasibility Assessment and Evaluated Criteria.....	17
Functional Criteria of MVP	17
Joint Torque Analysis and Motor Specification.....	18
Finite Element Analysis and Deflection Validation.....	18
Power Consumption and Battery Life.....	19
Intel RealSense Vision Validation	19
Experiment Definitions and User-in-the-loop Testing.....	19
Robot-only Testing	20
Final Design Solution	20
Risk Assessment, Safety and Liability	21
Societal, Environmental, and Sustainability Considerations	21
Conclusions and Future Work	21

Team Member Contributions.....	22
References	23
Appendices	26
Appendix A: Comparison of relevant predicate technologies.....	26
Appendix B: Stakeholder summary and key findings.....	27
Appendix C: Market Analysis & Growth	29
Appendix D: Function tree of critical design features and their subfunctions	30
Appendix E: Description of core device functionality and associated evaluation metrics.....	31
Appendix F: Engineering specification table for fully featured product (not MVP)	32
Appendix G: Engineering design standards applicable to SidewalkSidekick.....	34
Appendix H: Morphological chart for design concept ideation.....	36
Appendix I: Solution architecture concepts	37
Appendix J: Scoring guidelines used for concept down selection	38
Appendix K: Weighted downselection matrix showing quadruped architecture is most suitable for meeting customer needs.....	39
Appendix L: Risks and potential mitigation strategies associated with wheeled quadruped architecture	40
Appendix M: Preliminary node graph based on ROS 2 framework for MVP feasibility demonstration.....	41
Appendix N: User control flowchart for crosswalks.....	42
Appendix O: User control flowchart for curbs	43
Appendix P: User control flowchart for staircases and ramps	44
Appendix Q: Potential questions for stakeholder evaluation of handle prototypes	45
Appendix R: Feedback received from stakeholders and resulting design changes.....	46
Appendix S: Showcase of different gait paths of interest for Sidewalk Sidekick.....	47
Appendix T: Pinocchio simulation environment allowing for full inverse dynamic simulation to achieve relevant joint torques over walking and standing gaits.....	48
Appendix U: Relevant properties of 6061 T6 Aluminum used for manufacturing and SolidWorks FEA.....	49
Appendix V: FEA analyses of linkages of interest and main chassis	50
Appendix W: RealSense software architecture.....	51
Appendix X: Software nodes, their primary function, and libraries that are being used or explored	52
Appendix Y: Anthropomorphic measurements for handle design based off the ANSUR II database.....	53
Appendix Z: Societal Impact Assessment	54
Appendix AA: Handle Concept Design Pros and Cons for Down-Selection.....	56
Appendix AB: Feedback from first round user testing	57
Appendix AC: Time Spent to Identify Button Shapes on Updated Rigid Handle Prototype.....	58
Appendix AD: User preference components and framework.	59
Appendix AE: Detailed overview of second-round user testing (Experiment 1).....	61

Appendix AF: Estimated runtime for different states.	62
Appendix AG: MVP design criteria & evaluation results.....	63
Appendix AH: Robot-only obstacle avoidance testing results by obstacle size class.....	64
Appendix AI: Detailed overview of experiments	65
Appendix AJ: Tire design iteration	66
Appendix AK: Path planning algorithm iteration	67

Nomenclature and Glossary

BVIP: Blind and Visually Impaired People

CV: Computer Vision

DOF: Degrees of Freedom

FOV: Field of View

MVP: Minimum Viable Product

Orientation & Mobility (O&M): The specialized training, skills, and techniques required to safely and independently navigate environments by the visually impaired

Proprioception: the body's ability to sense its own position and movements

Support Phase: The part of the gait where a leg is in contact with the ground supporting the robot's weight.

Swing Phase: The part of the gait when a leg is in the air, moving forward, and preparing for the next contact.

Transient Barriers: Short-term, or movable obstacles in the environment that disrupt navigation, safety, and access.

Executive Summary

Navigating complex urban terrain is a challenge for blind and visually impaired people (BVIPs). Common obstacles include uneven sidewalks, curbs, construction zones, and transient barriers. These features pose significant safety issues for the BVIP community. While white canes and guide dogs remain the predominant technologies to assist with obstacle detection and navigation, limitations persist. These technologies fall short with regards to detection of non-ground-based obstacles and navigation to unfamiliar locations.

Through stakeholder discussions and market research, there is a clear demand for a user-friendly device that enhances independence among BVIP in urban environments. The objective of this project is to design an electromechanical guidance device that is capable of safely guiding BVIP through outdoor urban environments.

Both robotic and non-robotic solutions were explored during the ideation phase, including a wheeled rover with suspension, a quadrupedal robot, a transformable wheel rover, a modular acoustic-sensing cane, and a haptic feedback belt. These concepts were evaluated against weighted design criteria derived directly from 7 primary stakeholder needs: obstacle detection capability, terrain adaptability, guidance ability, ease of user interaction, robustness, unobtrusiveness, and total cost.

Based on this evaluation, the selected concept was a wheeled quadrupedal robot. This architecture combined the adaptability of legged locomotion with the efficiency of rolling motion. It also enables safe traversal of curbs, stairs, gravel, and uneven sidewalks while conserving power on flat terrain. The system also uses forward-facing vision sensors for early hazard detection, autonomous navigation and route execution, and a tactile interface that mimics familiar guide dog interaction. Together, this design increases independent mobility, reduces injury risk, and provides a practical alternative to current assistive technologies in urban locations.

Mechanical design was evaluated through Pinocchio simulation and FEA to support uneven-terrain navigation, while the electrical architecture fuses and split power distribution to reduce the risk of full system failure from a single component issue. Software testing supported the use of the Intel RealSense D435i for depth sensing and object tracking, and the prototype integrated early versions of obstacle detection, local rerouting, environmental description, handle-based input, and voice interaction. These functions were tested through controlled demonstrations intended to show how the robot could guide a user, respond to hazards, and communicate useful information during travel.

The prototype was developed to demonstrate the core feasibility of this final product concept. Stakeholder and user feedback played a central role in shaping the final design. Testing with a BVIP individual on campus enabled the evaluation of prototype feasibility for intuitiveness, comfort, and safety. This feedback reinforced that the device should not feel like a robot taking control away from the user. Instead, it should behave more like a guide partner by providing clear physical cues, communicating important hazards, and allowing the user to confirm or adjust actions when needed. As a result, the final design emphasizes shared control, improved handle ergonomics, tactile controls, open-ear audio feedback, and adjustable user preferences.

The final design is a shared control guidance system that includes a wheeled quadrupedal robot, a sensor suit, and intuitive feedback through proprioceptive & auditory communication. The user can select a destination through voice command or an interface, while the robot detects obstacles, plans around hazards, and guides the user through the handle. Future work includes expanding from local route demonstrations to full global navigation, integrating GPS, upgrading the sensor suite, improving the path planning algorithm, training perception models on a larger BVIP-specific hazard dataset, increasing battery life, improving weather resistance, and refining the device for reliability and small-batch manufacturability.

Introduction and Background

According to the Centers for Disease Control (CDC), over 1 million Americans are blind. Among this community, navigation remains a primary challenge in outdoor urban environments, as these settings frequently contain uneven terrain and transient barriers. Traditional mobility aids, such as white canes and guide dogs have limitations in detecting obstacles, providing real-time environmental feedback, and navigating unknown areas.

The objective of this project was to design an electromechanical guidance device that was capable of safely guiding BVIP through outdoor urban environments. Success was determined by the robot's ability to detect uneven terrain and obstacles from ground to head level, identify hazards up to five meters ahead, use grid-based mapping algorithms to autonomously plan and execute a path, and provide intuitive user feedback through a handle interface and web application. Another key criteria was that the device must also remain compact, durable, cost-effective, and have sufficient battery life to operate for the entirety of the user's day. Additionally, it was important to design a device that mimicked frameworks already used by visually impaired individuals as opposed to forcing them to adapt to unfamiliar technologies. In these environments, existing mobility solutions do not fully support independent navigation for visually impaired individuals. Guide dogs provide effective assistance but have inherent limitations, including unidirectional communication. Other technologies often rely on the user having to use multiple tools simultaneously, such as a white cane, several smartphone applications, wearables, and occasional sighted assistance. Conversations with stakeholders showed that this combination of tools still does not provide enough spatial context with hazard detection; there is a desire for a single device that would combine and improve upon all these functions.

There was a clear need for innovation in low-vision assistance technology. The following sections describe the evaluation approach, compare leading concepts against the project criteria, and explain the rationale for selecting a quadrupedal robotic architecture for the project concept.

Existing Products, Prior Art, and Relevant Literature

To understand the concept selection process, it was necessary to examine existing and emerging assistive technologies available to BVIPs. Multiple institutions have reviewed these technologies to identify persistent limitations of navigation for BVIP [1][2][3]. Mature and widely adopted solutions, such as the white cane, primarily function as short-range detection tools [1]. While the white cane is low-cost and highly accessible, it requires the user to constantly interact with the environment, thereby reducing ambulatory speed. These constraints conflict with project objectives, as decreased independence is associated with lower ambulatory speeds. Another established solution is the use of guide dogs, which support navigation by maintaining a consistent spatial relationship between the human and the dog [1]. Through this interaction, the guide dog provides a proprioceptive reference that assists the user in maintaining orientation and perceiving changes in ground topography. However, guide dogs cannot audibly convey environmental information to the user and are difficult to provide at scale due to cost and availability constraints.

To either supplement or replace these mature concepts, newer technologies have been explored in recent years. These concepts typically fall into two categories: wearables and robotic systems. A recent survey of wearable navigation technologies explores three different subcategories based on where computation occurs: environmental beacons, in the cloud, or on the wearable devices themselves [2]. Further, wearable devices can be positioned at different body locations, with current products such as Maptic, worn on the neck or wrist and GUIDi, designed for use on the waist. These systems consistently rely on non-proprioceptive feedback, delivered through auditory or haptic cues to guide the user through their environment. This limits the intuitiveness of these devices, as the user must continuously interpret abstract signals and may not be able to physically localize obstacles in an absolute sense. When paired with a white cane, wearables can improve awareness of upcoming obstacles and increase detection rates [4]. However, the speed limitations associated with physically locating obstacles using the cane remain. Further, using multiple assistive technologies may increase cognitive load for the user. As a result,

wearable technologies exhibit drawbacks in cluttered environments where rapid decision-making and continuous obstacle tracking are required.

On the other end of emerging technologies are robotic concepts [3]. Robotic navigation systems typically provide proprioceptive feedback through physical coupling, producing a “tug” or guiding force that can more intuitively direct users through the environment. These platforms may even be supplemented with auditory or haptic feedback to improve user experience. Commercially available solutions, such as the wheeled robot cane Glide, address several challenges associated with urban navigation but still require the user to lift or manually manipulate the device to traverse transient barriers. To address uneven terrain more directly, researchers have investigated quadrupedal robots intended to replicate aspects of guide dog locomotion [5]. However, a recent study indicates that user trust in robotic guide dogs can be lower than trust for wheeled platforms [6]. Participants cited locomotion noise and increased proprioceptive degrees of freedom, such as lateral leg motion, as contributing factors to discomfort and distrust. Additionally, surveys of these platforms often do not emphasize the direct feedback mechanisms provided to the user, as many robotic dogs lack the explicit communication interfaces commonly used in wearable systems, which can further increase cognitive load during navigation. Relevant predicate technologies are summarized in Appendix A, including product examples, technical comparisons, and feasibility for the problem objective.

Stakeholders and Market Research

Stakeholder groups were identified and engaged through interviews, focus groups, and site visits to capture both lived experience and expert insight, as seen in Table 1. Further details on stakeholder conversations, market research, and key findings are in Appendices B and C.

Table 1. Stakeholder type, number, affiliation, and engagement method

<i>Stakeholder Type</i>	<i>Affiliation</i>	<i>Engagement Method</i>
<i>Visually Impaired Individuals (3)</i>	Shepherd Center, Georgia Tech (Able Alliance)	User/Expert Interview, Stakeholder Testing
<i>Occupational Therapist (1)</i>	Center for the Visually Impaired (CVI)	Interview/Focus Group
<i>Assistive Technology Specialists (3)</i>	Tools for Life	Interview & Site Visit
<i>Research Specialist (1)</i>	Mississippi State University NRTC	Expert Interview
<i>O&M Specialist (1)</i>	Center for the Visually Impaired	Expert Interview

Customer Requirements and Engineering Design Specifications

Conversations with stakeholders and analysis of predicate technology were directly translated into customer requirements and corresponding engineering specifications. As a result, a key gap in assistive technology for people with visual impairments was identified – current solutions either take individuals from point-to-point or provide a narrow level of environmental contextualization. While each is an essential component of urban navigation, enabling safe travel to unfamiliar destinations necessitates an integrated platform for real-time, preventative barrier detection.

Criteria Table

To guide the design of this integrated mobility platform, a set of criteria was derived from customer needs. Aside from framing the problem space, Table 2 also serves as a communication tool between stakeholders and the engineering team. By outlining high-level and solution agnostic functionality, the criteria table keeps teams aligned and provides a pathway for continuous user-centric development critical to assistive technology.

Table 2. Criteria Table: Sidewalk Sidekick

Enhances independent mobility for individuals with visual impairment in outdoor urban environments
--

Detects hazards at all levels, including uneven terrain and obstacles from ground to head-level	Intended for use on sidewalks, crosswalks, stairs, and shared paths
Detects hazards far ahead of user (up to 5 m)	Operates at ambulatory speeds
Navigates user to new locations	Minimizes cognitive load required during use
Alerts user with sufficient time to respond	Supports multimodal communication based on user preferences
Maintains a compact, unobtrusive form factor	Lasts for multiple round-trip journeys on single charge
Minimizes shoulder, wrist, and arm strain during extended use	Functional in normal weather conditions (water resistant)
Can be easily carried and transported when indoors or in vehicles	Affordable and competitive price-point (<\$60K) compared to the cost of training and taking care of a guide dog (\$60,000 to train and \$180 - \$200 a month to take care of it)

The defined criteria prioritized safe and independent navigation in unfamiliar urban environments, emphasizing proactive detection of terrain-based hazards and barriers not detectable via a traditional white cane. Additional criteria focused on the usability and accessibility of the design by minimizing cognitive and physical strain through multimodal communication and comfortable interfaces. Social integration was another consideration, ensuring that the solution maintains a compact, unobtrusive form factor for BVIP. Cost and reliability guidelines were also established to build trust with end-users.

Key Functionality

Project criteria defined what the device must achieve, but they were limited in describing how the system must operate to meet these needs. A function tree was used to ensure all requirements were met, providing a top-down framework for establishing core features of a proposed solution. The five main functional categories and their relevant subfunctions are defined in Appendices D & E.

Engineering Specifications

Base-level functions defined in Appendix D informed the development of engineering specifications required to meet BVIP needs. Table 3 captures key technical specifications and justification for quantitative and qualitative data based on standards and operating conditions.

Table 3. Engineering Specifications

<i>Functional Category</i>	<i>Demand vs Wish</i>	<i>Technical Requirement</i>	<i>Rationale/Source</i>	<i>Priority</i>
<i>Traverse Outdoor Urban Terrain</i>	D	Maintain stable operation on terrain with local surface height variations up to 38 mm (1.5 in)	Surface roughness beyond this threshold is considered hazardous for individuals with disabilities – should be avoided via rerouting [13]	High
	D	Able to traverse vertical discontinuities between 100-240 mm (4-9.5 in)	OSHA minimum curb height and maximum stair riser height [14]	High
	D	Achieve minimum turning radius of ≤ 0.6 m	Enables 90° turns while remaining centered in the smallest width sidewalk of 1.2 m [15]	High
	W	Support operating speeds up to ≥ 1.24 m/s	Mean adult walking speed reported in pedestrian mobility study [16]	Medium
	D	Ingress Protection (IP) rating IP54	Required for outdoor urban operation [IEC 60529]	High
<i>Detect Permanent & Transient Barriers</i>	D	Identify terrain height variations or gaps ≥ 6.35 mm (1/4")	OSHA tripping hazard minimum height [17]	High
	D	Detect obstacles within 0-5 m range	Provides ~4-5 seconds of lead time at walking speeds	High
	D	Detect obstacles from ground level to 2 m (6'6")	Accommodates head-level obstacles for 99% male height	High
	D	Perception latency of ≤ 250 ms	Maintains ability for average user to respond to obstacles as close as 50 cm	Medium

<i>Enable User Control & Interaction</i>	D	Provide configurable auditory and tactile feedback	Driven by user preferences	High
	D	Voice command recognition with up to 73.4 dBA background noise	Study of mean urban street noise levels [18]	Medium
<i>Guide & Navigate Users to New Destinations</i>	D	Average absolute localization accuracy of 5 m	Derived from current GPS technology accuracy	High
	D	Relative localization accuracy of ≤ 30 cm	Enables sidewalk positioning and obstacle avoidance in at least 4 zones	High
<i>Support Community Data Sharing</i>	W	Capable of storing and retrieving hazard information from a geotagged database	Shared experience and first-hand knowledge pool between users	Low
	W	Support capture and storage of video (≥ 3 min, ~ 1.5 GB) for emergency situations	Occasional need for later reference or sighted individual guidance	Low
<i>Packaging</i>	D	Device volume $\leq 350 \times 350 \times 350$ mm	Allows for safe use on walkways, enabling pedestrian passing	High
	D	Total weight ≤ 6.5 kg	NIOSH lifting guidelines and safety factor of 1.25	High
<i>Cost</i>	W	Total BOM $\leq \$1000$	Maintains ability to price as a low-cost assistive technology	Low

Relative priorities were derived from traceability studies between customer needs and engineering specifications. Greater weight was placed on requirements that support multiple device functions.

Codes and Standards

Relevant codes and engineering design standards for SidewalkSidekick are in Appendix G.

Design Evolution

Concept Generation

The criteria table determined from stakeholder meetings and market research directly informed the construction of the morphological chart (Appendix H), which organizes concept alternatives for achieving each key device function: navigation of outdoor urban terrain, detection of permanent and transient barriers, ease of user interaction, guidance to new locations, and communication between users. From this, the concept ideation process was first divided into two primary terrain navigation categories: robotic mobility platforms and wearable aids. Robotic concepts included a wheeled rover, quadruped robot, and a transformable wheel rover – all of which focused on providing proprioceptive guidance and user-defined autonomy in urban environments. Wearable and modular concepts included an acoustic-sensing cane and a navigation belt, both of which emphasized compact design and are inspired by proven and familiar assistive technology. Perception and user interface concepts were considered across multiple locomotion and non-robotic platforms because every viable solution required methods for environmental sensing, obstacle interpretation, communicating information, and receiving user input. This allowed for comparisons between high level architectural differences, while still carrying forward the most effective sensing and interaction ideas from the brainstorming phase into a selected concept. The resulting solution architectures and their tradeoffs are summarized in Appendix I.

Concept Selection

All concepts were evaluated using the scoring guidelines in Appendix J and the weighted decision matrix in Appendix K. Seven functional categories were selected to represent the most important customer needs during down selection. Based on this evaluation, the quadrupedal architecture was selected, as it provides an integrated solution to the problem objective, combining positive features of the white cane, guide dogs, and apps that rely on human agents. Prior research shows that quadruped locomotion is an ideal solution for climbing stairs, driving across curbs, and traveling on uneven terrain due to its adaptive gait control and high ground clearance [19]. The inherent adjustability of sensor positioning results in a larger FOV,

allowing the device to detect objects from farther away when compared with alternate and existing solutions. It also allows for obstacle detection and classification from ground level to head level, in line with requirements in Table 3. In addition to obstacle detection capability, the device can navigate the user around obstacles, like a guide dog but requiring less post-travel care. Additionally, prioritizing space-efficient and modular mechanical design during the initial design phases will allow the device to be easily transported and stored in places like airplanes, vehicles, and restaurants.

Although the quadruped architecture best fulfills the design criteria, there are several potential limitations that are worth discussing. A traditional, high-DOF-quartered architecture requires twelve actuators, adding an additional layer of control complexity. The control algorithm must also incorporate dynamic force compensation to allow for a continuous transition between the swing phase and support phase during locomotion, as each phase comes with varying speed, force, and torque requirements [20]. The four-bar leg linkages also increase component count across the robot, increasing potential failure points. Legged motion is less efficient than rolling motion due to complex, energy intensive movements that are required to lift, swing, and place limbs [21]. This results in higher power consumption and directly conflicts with the design criteria, which prioritizes a long battery life. Therefore, to decrease overall power consumption and control complexity, the device was designed to incorporate a wheeled end effector on the leg rather than the semi-compliant ball end effector seen on other quadruped architectures. The wheels offer predictable kinematics that simplify navigation, creating a more stable base for the robot. In addition, studies have shown that BVIP trust wheeled robots more as guidance devices because the lack of lateral rocking motion that accompanies a traditional quadruped gait translates as increased navigation confidence [5]. However, when legged locomotion is needed for more discrete barriers such as curbs, branches, and scooters, the system can operate and successfully navigate them.

Concept Iteration

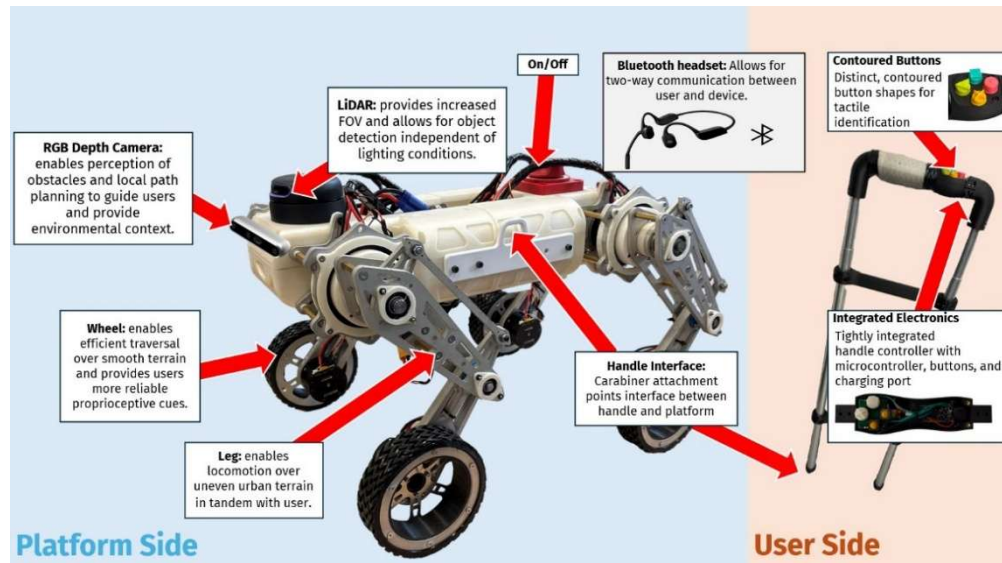


Figure 1: Representative drawing of final prototype.

Selection of the wheeled quadruped architecture marked a shift from brainstorming to subsystem-level design. While the core robotic locomotion platform remained largely unchanged throughout development, the final prototype's mechanical, electrical, and software implementations evolved with a focus on improving human-robot interaction as testing revealed integration constraints and additional user needs.

A representative drawing of the final prototype is shown in Figure 1. The following subsystem overviews detail the mechanical, electrical, software, and user-interface implementations, while highlighting the major iterations shaping the final design.

Mechanical Architecture

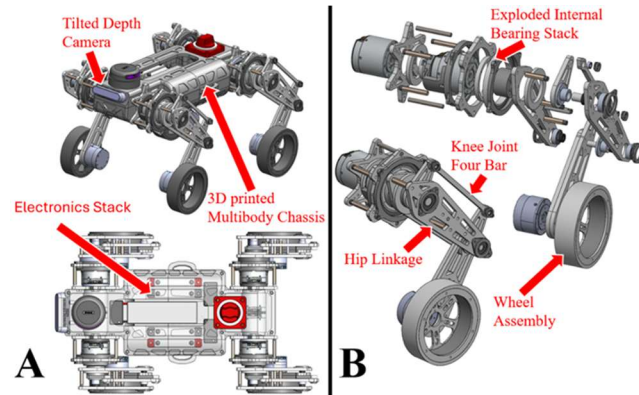


Figure 2: Overview of relevant mechanical systems. A) showcases the overview of the entire assembly and the chassis. B) overviews the individual leg subsystems.

As previously mentioned, a wheeled quadruped was selected, with rolling as the primary mode of locomotion, the legs acting as suspension, and walking used for specialized, discretized barriers. An overview of the mechanical assembly is shown in Fig. 2. This distinguishes the system into two primary components: the chassis and the leg. Beginning with the latter, the leg design for Sidewalk Sidekick incorporates a four-bar linkage that is coaxially defined with the first hip motor. However, this mechanism is not arranged in a parallel configuration and is instead more like a serial architecture, eliminating the need for differential control. The design is supported by an external bearing stack, which minimizes deflection and enables high precision of

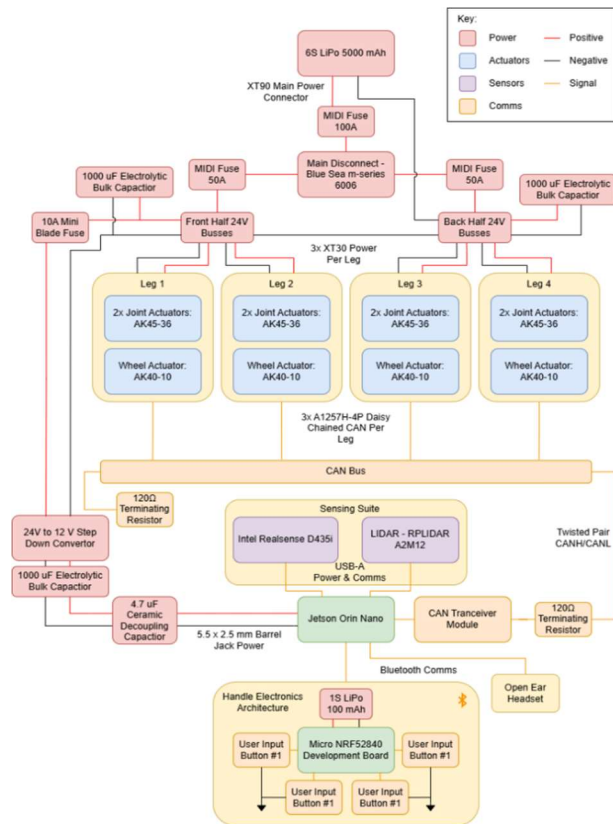
the end effector against the ground. Material choices and the ability to dual support the leg shafts were intentionally selected to provide precision, allowing the user to accurately feel the dynamics of the platform without excessive play. For this reason, metals such as aluminum 6061 were used in key load-bearing areas to meet structural requirements. In contrast, the chassis was constructed from plastic with wall thicknesses exceeding 0.25 in and optimized geometry to withstand torsional and bending loads while maintaining a lightweight design. Reduced weight leads to more favorable torque requirements, with further analysis of these tradeoffs discussed in the engineering analysis and FEA sections. Additionally, the use of a custom 3D-printed chassis allowed for precise control over geometry and shape, enabling tightly integrated electronics and contributing to a more ergonomic and familiar user experience.

For the robotic platform, mechanical iterations focused on improving the smoothness and predictability of proprioceptive feedback delivered to users. Testing with the initial tire design (Appendix AJ) revealed noticeable vibrations and inconsistent rotational behavior during turns. The first tire design had a large contact patch and a directional tread pattern. Because the robot relies on skid-steer motion, the wheels must partially slip during rotation; with the contact patch spanning the tire's full width, there was increased friction and tire scrubbing, making it difficult for the robot to pivot smoothly. The directional treads on the first tire introduced traction asymmetries between forward and reverse wheel rotation, which made it harder for the platform to rotate about its center. These effects made guidance less intuitive and predictable. The revised tire design, shown in Appendix AJ, uses a reduced contact path and a circumferential tread pattern to improve turning smoothness, reduce tire skipping, and provide more predictable turning dynamics for users.

Electrical Architecture

The electrical architecture of SidewalkSidekick incorporates components with simple interfaces, when possible, to reduce integration risks. Four key stakeholder criteria strongly influenced this design: maintaining a compact form factor, minimizing cognitive load during use, reducing fatigue during extended operation, and supporting use over multiple round-trip journeys on a single charge. Although battery life is less important for MVP feasibility, the design keeps this in mind.

Downstream of the disconnect, power is split between two distribution buses that are each protected by a 50 A fuse. This choice serves two main purposes: ease of packaging in a smaller chassis and, more importantly, fault isolation due to unexpected failures. The electrical architecture originally included a step-down convertor to supply compute and sensing hardware from the main actuator battery, along with additional bus-level capacitance to protect against voltage sag. During integration, this architecture was simplified. The final prototype uses a separate 4S LiPo battery for the Jeston and connected sensing, electrically isolating compute and communication from actuator noise. This change reduces the risk of dynamic motor current draw interrupting perception, audio feedback, or high-level control. Testing also showed that the 6S actuator battery had high discharge capability, so the bulk capacitance on each bus was removed to reduce system complexity.



Most importantly to the user, the handle electronics architecture focuses on providing an intuitive interface that minimizes fatigue during use. By selecting a low-power Bluetooth MCU, and a small 1S LiPo power supply, the handle remains extremely lightweight. This MCU supports necessary button inputs, while more complex interactions are handled through voice commands using the OTS headset. The handle hardware also features integrated USB-C charging for a more user-friendly design.

Software Architecture

Software is the final subsystem required for SidewalkSidekick to serve as an integrated platform for navigation and contextualized guidance for BVIP. MVP evaluation is focused on aspects of the device not captured by predicate technology. This is primarily the user interface and the platform's effectiveness in navigating specific urban scenarios that are commonly challenging for visually impaired people. These objectives can be evaluated without implementing full autonomous navigation, and this idea heavily influences the software choices.

ROS 2 was selected as the original framework for two main reasons: its existing packages for robotic applications, and its modular architecture that scales beyond MVP delivery. The robot's functionality is decomposed into nodes that manage specific functions and communicate information with other nodes through topics, actions, or services. Topics are used for continuously shared data, like sensor streams and status variables. Actions are used for behaviors that take time to complete, such as executing a route. Services are used for short requests with immediate responses, such as enabling a microphone for command transcription. A more detailed diagram highlighting the connection between each node and ROS 2 specific details can be found in Appendix M. However, the final prototype software was implemented as a lower-overhead Python package to reduce integration complexity during MVP testing. This iteration preserved the functional decomposition originally planned and can easily be extended into a ROS 2 implementation for future development. Appendix X outlines the Python architecture's primary software components implemented in the feasibility prototype and their respective functions. The system is grouped into modules for human-robot interaction, perception and path planning, and motor control and state estimation. Together, these layers bridge the gap in existing assistive technology for BVIP by providing a platform that behaves more like a sighted guide through sensing, adaptive locomotion, and communication.

Software iteration focused on improving path planning reliability and human-robot interaction strategies. Initially, obstacle avoidance used global replanning through an adaptive A* approach. This algorithm projected detected obstacles into a global occupancy grid and replanned a route around them. While this implementation was originally selected to improve path efficiency and reduce travel time for the user, it proved challenging to integrate reliably. Because detected objects were transformed into a global frame, small errors in pose estimation accumulated over time and reduced the fidelity of obstacle placement within the map. Despite the use of sensor fusion between IMU streams and odometry data for pose estimation, inherent noise in these sensors made reduced global pose accuracy. This produced inconsistent obstacle representations in the global grid, as shown in Appendix AK, where the true obstacle is represented by the blue square and the detections projected into the global grid are shown in red. Since detection accuracy, route completion, and user safety were more important to MVP feasibility evaluation than globally optimal path planning, the navigation strategy was simplified to a local costmap obstacle avoidance approach. In this approach, depth information from the RealSense is filtered into a local occupancy grid rather than projected into a global map. This improves detection fidelity because the robot makes decisions based on obstacles currently detected in its local frame instead of relying on a persistent global obstacle representation. A visualization of the local costmap is shown in Appendix AK. The motion planner selects a feasible direction that maintains forward progress while avoiding nearby obstacles. This change better matched the MVP objective by improving responsiveness, reducing oscillatory behavior, and focusing on safe local avoidance rather than optimal route planning. Further perception improvements were made by filtering depth information and simplifying direction-selection heuristics. These changes reduced the time from obstacle detection to motion decision from

approximately 2400 ms to 20 ms, enabling a more responsive platform for user guidance. Software defining human-robot interaction was also continuously improved through testing. For example, after testing, velocity commands were smoothed upon startup and shutdown to reduce shoulder strain from sudden acceleration and support natural human motion. Voice command functionality was also expanded to include incremental speed adjustments to support different walking preferences.

Human-Robot Interface

The handle served as the primary interface between the user and the robot. This connection shaped user interaction and information delivery, conveying control, feedback, and navigation through tactile and proprioceptive interaction. The design integrated industrial design decisions in conjunction with human factors principles to deliver a safe, intuitive, and low-effort user experience for BVIPs. Key considerations for design included reducing muscular strain, minimizing cognitive load, and balancing autonomous decision making with user control to provide a simple and safe platform. Success in these criteria was governed by anthropometry and tactile and visual interaction in mind.

Anthropometric and Ergonomic Considerations

The length and adjustability of the handle connection to the device was designed to accommodate a wide range of users. In the United States, average adult heights range from approximately 150 cm (5% female) to 180 cm (95% male). Keeping this height range in mind during the design process ensures the device is usable by the majority of BVIPs. Additionally, this ensures the device can be customized to ensure it meets height requirements provided by the National Federation for the Blind. Dimensions of the handle found in Appendix Y are determined using the ANSUR database (Anthropometric Survey of U.S. Army Personnel) [23]. When designing the handle and its attachment to the guidance device, adjustable height mechanisms enable users to maintain a neutral wrist posture (0-15° extension) to reduce musculoskeletal strain during prolonged use. In addition, a D-pad style button configuration further reduces the range of thumb motion required.

Tactile and Visual Interaction

Given the unreliability of voice interaction in urban environments, the handle also integrated tactile-first interaction through physical buttons. These buttons were guided by human factors principles and stakeholder preferences. Buttons utilized 3D shapes and universal symbols to allow identification without visual confirmation. Additionally, BVIP is a spectrum with some individuals having a low-medium visual impairment. While color is never the sole differentiator, buttons are designed to be brightly colored to provide additional guidance for individuals with these impairments. The bright hues of the buttons therefore create redundancy in the button identification process. Collectively, these features reduce cognitive load by enabling recognition-based interaction as opposed to recall, which is important for visuals who are navigating through complex outdoor environments.

Concepts

With these considerations, stakeholder feedback, and literature-based research, three concepts were initially explored: Retractable Dog Leash Handle, Rigid Handle, and Pull-Strap Handle. A comparative summary can be found in Appendix AA. After discussing with stakeholders, the retractable dog leash handle was eliminated from consideration, as it had only one point of central contact with the robot. This prevented the user from having a clear understanding of robot position when in use. Thus, the two concepts that required two points of contact around the body of the robot (rigid handle and pull strap handle) were tested with a Raveena Ali, a blind individual on campus and subsequently iterated upon based on feedback.

User-Interface Testing and Validation

Raveena assessed the rigid and pull strap handle designs for comfort and functionality. Raveena was asked to walk with the mobility device for 10 meters, with randomized stopping and starting motions along the path and was asked the questions found in Appendix Q. Her responses and feedback were used to downselect between handle styles. Criteria for evaluation included effective communication & responsiveness, secure grip and posture, tactile & visual button identifiability, and familiarity & low cognitive load. Initial handle prototypes demonstrated a tradeoff between proprioceptive feedback and overall comfort and grip security (Appendix AB).

Proprioceptive Feedback: When using the pull strap handle, there was a visible lag between when the robot stopped vs. when Raveena interpreted the motion and stopped herself. This was due to the lack of rigidity in the flexible handle causing it to buckle. Switching to the rigid handle, Raveena was able to easily perceive the robots starting and stopping motions, and a clear response time reduction from ~540 ms to ~100 ms. Raveena also reported a clearer understanding of robot motion with the rigid handle.

User-Interface Feedback: Raveena evaluated grip security, comfort, and button identification for each handle concept during use with the robot. Concerns were expressed regarding the diameter of the rigid handle, as Raveena had trouble wrapping her hand around the handle. Additionally, when asked to discern the button shapes feel alone for both handles, she was correctly able to identify three out of the four shapes in under 1 minute; the “describe surroundings” oval button was unidentifiable.

Final Handle Design: The rigid handle was selected for further development due to its improved proprioceptive feedback that makes guidance safer and more intuitive. Incorporating learnings from early testing, several changes were made to the design. First, the handle diameter was reduced from 48 mm to 34 mm to enable a more comfortable closed grip during use. Additionally, to enhance the grip on the handle, a custom VytaFlex urethane polymer mold was cast and fit to the body of the handle. This provided an extra layer of grip support and padding. The texture difference between the urethane sleeve and the buttons also aided in the identification of button control placement. Since the oval test button was difficult to distinguish, it was redesigned to be an “X” for easier identification. Tactile bumps were also added to the corners of each button to add a layer of redundancy. Additionally, buttons were raised to be 3mm in height, and the spring-loaded pin design of the rigid handle was designed to be adjustable from 71cm to 86 cm in length to accommodate the average adult height range. With these new changes, Raveena was able to identify the button shapes within 30 seconds. The effectiveness of these changes was also tested on a random sample of 10 students on campus. When asked to identify the shapes of the buttons on feel alone, the average time spent identifying the buttons was 29.9 seconds. Test results can be found in Appendix AC. The final handle design is shown in Figure 4 and final button functionality is detailed in Table 4.

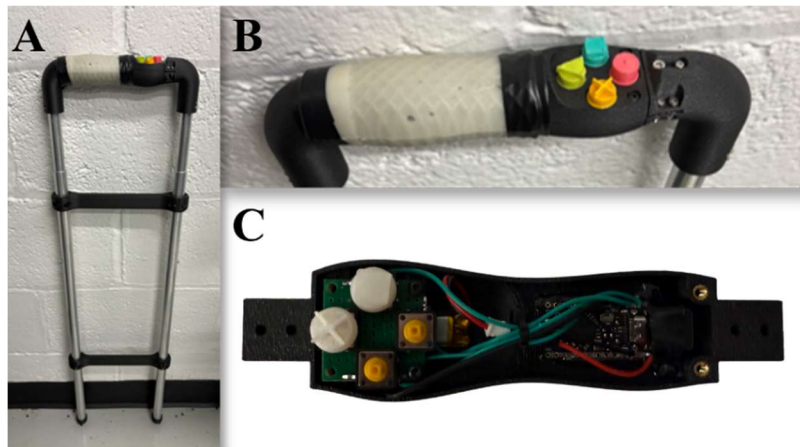


Figure 4: Final handle design. **A** showcase entire design. **B** showcases top handle section. **C** showcases the internal handle wiring.

Table 4. Buttons and their respective functions

<i>Button</i>	<i>Function</i>	<i>Appearance Description</i>
1	Stop Movement	Red (#ff0000) 3mm raised circle
2	Describe Surroundings	Yellow (#ffff00) 3mm raised X button
3	Start Movement	Green (#00ff00) 3mm raised triangle
4	Touch to Talk	Blue (#0000ff) 3mm raised square

Feasibility Assessment and Evaluated Criteria

Functional Criteria of MVP

To prioritize the development of user interaction and demonstrate feasibility of an MVP during capstone expo, evaluation is limited to a subset of criteria derived from full engineering specifications. The criteria in Table 5 were selected to validate the key features of SidewalkSidekick that differentiate it from existing assistive technology and fill the gaps expressed by BVIP. Rather than demonstrating autonomous navigation, the MVP focuses on developing the assistive framework and evaluating whether the shared control between user and robotic platform work to enhance independence.

Table 5. Evaluated Criteria for MVP Delivery

<i>Criteria from Table 2</i>	<i>MVP Evaluation</i>	<i>Final Product Evaluation</i>
<i>Detects hazards at all levels, including uneven terrain and obstacles from head to ground level</i>	Must detect relevant stationary hazards in front of the user from ground level up to 6 ft 6 in. Does not need to detect moving obstacles or provide full 360° awareness.	Detects both stationary and moving hazards across the full required height range and provides 360° awareness for more fully autonomous operation.
<i>Detects hazards far ahead of user (up to 5m)</i>	Detects hazards up to 3 m ahead and as close as 1.6 m in front of the guidance device.	Detects hazards up to 5 m ahead and as close as 0.5 m, including hazards requiring dynamic response.
<i>Navigates user to new locations</i>	Provides guided assistance on a preset mock urban route with variable obstacle placement. Does not need to navigate to new destinations using GPS, mapping, or world-model localization.	Navigates users to new destinations in unstructured urban environments; includes an interface for destination selection.
<i>Alerts user with sufficient time to respond</i>	Must alert the user of hazards at least 2 seconds before they reach the hazard location.	Capable of providing users with up to 5 seconds of lead time before reaching hazard location.
<i>Maintains a compact, unobtrusive form factor</i>	Volume $\leq 350 \times 400 \times 350$ mm (L x W x H).	Volume $\leq 350 \times 350 \times 350$ mm.
<i>Minimizes shoulder, wrist, and arm strain during extended use</i>	Tethering platform and handle design conform to existing interaction strategies familiar to guide dog users.	Same as MVP.
<i>Can be easily carried and transported when in indoor areas or vehicles</i>	Device weight < 15 kgf.	Same as MVP
<i>Intended for use on sidewalks, crosswalks, and shared paths</i>	Device operates in controlled environments designed to demonstrate feasibility in representative urban scenarios.	Operates in unstructured urban environments and is specifically designed for sidewalks, crosswalks, and shared pedestrian paths.
<i>Operates at ambulatory speeds</i>	Device supports motion of at least 1.24 m/s (mechanical speed)	Same as MVP.
<i>Minimizes cognitive load required during use</i>	Showcases different levels for control and feedback during guidance through independent demos.	Enables user definition for level of control and feedback during guidance through app-based selection.
<i>Supports multimodal communication based on user preferences</i>	Includes proprioceptive and auditory cues during route guidance. Includes buttons for user interaction and limited set of voice commands for demos.	Retains MVP communication avenues, with expanded voice command capabilities and destination selection feature.

<i>Lasts for multiple round-trip journeys on a single charge</i>	Battery life sufficient to support planned demos. Does not need to support repeated trips.	Supports multiple round-trip guidance sessions on a single charge.
<i>Functional in normal weather conditions (water resistant)</i>	Waterproofing and dustproofing are not required for feasibility evaluation.	Device designed to meet IP54 specifications.
<i>Affordable and competitive price-point compared to similar fidelity assistive technologies</i>	MVP prototype cost estimate is not expected to be representative of fully featured solution.	Final product is expected to cost significantly less than the initial burden of a guide dog, which is around \$60,000. Should be eligible for government and donation-based assistance pathways that already exist for guide dogs to reduce burden on BVIP.

Joint Torque Analysis and Motor Specification

The most critical analysis involved determining the inverse dynamic torques for the quadrupedal actuators, since poor selection would prevent the robot from achieving reliable and dynamic articulation. This process was carried out using the Pinocchio simulation environment with Meshcat as a visualizer. A first-order CAD-based URDF model was generated and used to simulate link inertias and masses in order to compute full rigid-body torques. A conservative 15 kg robot mass was assumed, with linkage lengths of 140 mm and 120 mm to enable traversal over obstacles greater than 6 inches, such as curbs and stairs. Relevant gaits were then analyzed, as shown in Appendix S. To evaluate these gaits, joint angles, velocities, and accelerations were prescribed at both the hip and knee for a trotting gait with a period of 0.2 seconds, during which two legs are off the ground at any given time. This analysis resulted in maximum RMS torques of 6.37 N·m at the hip and 7.24 N·m at the knee. Rolling RMS torques were lower and were therefore not considered when specifying the actuator. Torque estimates for a single leg stepping over a standard curb or stair showed that dynamic walking produces higher peak torques than static climbing. Therefore, actuators were sized for dynamic locomotion rather than curb-climbing. As a result, the robot climbs curbs one leg at a time, reducing actuator cost and complexity. Accounting for peak dynamic loading and applying a factor of safety of 2, the required torques increase to 21.4 N·m at the hip and 22.3 N·m at the knee. This process is summarized in Appendix T, although dynamic loading and the applied factor of safety are not shown in the output graph. These requirements make the CubeMars AK45-36 PMSM actuator a suitable choice, as it supports impedance control while meeting the necessary torque demands. Additionally, the CubeMars AK40-10 provides an appropriate balance between torque and speed, enabling ambulatory velocities of up to 1.24 m/s when paired with a wheel diameter of 125 mm.

Finite Element Analysis and Deflection Validation

With inverse dynamics complete, bearing reaction loads and joint torques were obtained for the first-order model. These values were then applied to the refined leg prototype used for the MVP. The primary objective of these analyses was to determine whether deflection at the end effector, the wheel, remained within acceptable limits for reliable traversal. The analysis led to reinforcement of the knee joint, as shown in Appendix V, where initial results indicated a deflection of 1.0 mm at the end of the knee. An additional 0.25 inch 6061-T6 aluminum plate, with properties shown in Appendix V, was added for support, reducing the deflection to 0.03 mm at the wheel. This process was repeated for all relevant linkages and ensured that the design met the requirements for user proprioceptive feedback, as the robot exhibits minimal mechanical play during traversal. Further analysis of the chassis was conducted to determine whether plastic or metal prototypes could withstand the forces and moments transmitted from the leg subassembly. A simplified box model was first developed to evaluate material feasibility. If a metal chassis were required, the increased mass would raise joint torque demands, which is undesirable. Preliminary results indicated that plastic could achieve deflections below 0.003 mm when designed with an appropriate wall thickness of 0.5 inches and geometries favorable for resisting torsional loads. Based on these findings, a chassis design was selected, as shown in Appendix V.

Power Consumption and Battery Life

To ensure that the platform can sustain a long operational lifetime for both the capstone expo and user testing, a preliminary battery life study was conducted for the actuator power system. The initial power architecture used a single LiPo battery with a buck converter to supply both the actuators and the Jetson. However, the final design employs a separate LiPo battery for the Jetson and LiDAR, which provides a runtime exceeding 5 hours. As a result, the actuator power system becomes the limiting factor for operational lifetime. Using the previously determined joint torques for both walking and rolling gaits, a lifetime analysis was performed under the assumption of continuous operation in each mode. Since real operation will not involve constant motion, this provides a conservative lower bound on battery life. Results were obtained using CubeMars current curves based on the PMSM torque–current relationship and a 5000 mAh LiPo battery, as shown in Appendix AF.

Intel RealSense Vision Validation

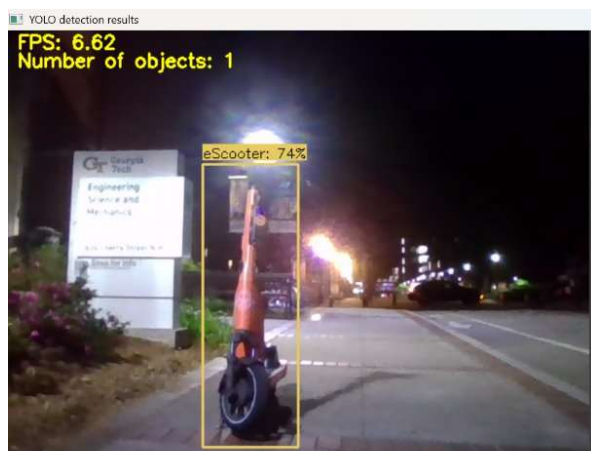


Figure 5: RealSense camera view showing scooter.

The perception architecture uses a YOLO (“You Only Look Once”) based object detector running on the RGB camera in the Intel RealSense D435i. YOLOv11 was used as the model for its optimized speed and accuracy over alternatives like R-CNN detectors that prioritize accuracy at the cost of additional computational overhead and latency. Specific Python scripts used to deploy this pipeline are included in Appendix W along with brief descriptions of each’s function. To improve obstacle classification performance for the specific case of visually impaired navigation, the *Image Dataset Comprising Objects and Obstacles in Outdoor Mobility of People with Visual Impairment* was used to train a custom YOLO model. This dataset contains classes pertaining to common

urban sidewalk hazards for BVIP, such as e-scooters, benches, and poles. As shown in Figure 5, the trained model can identify e-scooters in camera view. This demonstrates that the YOLO detector is sufficient for initial real-time scene interpretation and obstacle awareness using the RealSense RGB stream. Depth accuracy was evaluated by comparing the depth reported by the RealSense stereo depth camera script to a manual ground-truth measurement of a fire hydrant in the test environment. The estimated depth differed from manual measurement by approximately 4%, indicating that the system provides reasonably accurate distance information for the prototype. The camera FOV also achieves the required parameter of displaying 6 feet vertical height from a 6-foot horizontal distance, more accurately displaying 6’6” height detection from 1.93 meters.

Experiment Definitions and User-in-the-loop Testing

The following experiments were defined to establish the functionality of the MVP and feasibility of the final product. Since replicating the full complexity of outdoor urban navigation is outside of the given timeline, these demonstrations showcase the interactions between perception, locomotion, and user communication in mock environments to characterize the user experience and usefulness of the device. A summary of each experiment is located below, and a detailed description of independent, dependent, and controlled variables can be found in Appendix AI.

Experiment 1 (Stationary Object Detection and Avoidance): This experiment focused on detection & avoidance of obstacles and relevant communication to the user. *Experiment 2 (Shared-Control Guidance and Turn Execution):* This experiment will assess how effectively the robot and user collaborate to navigate turns using shared control and verbal interaction. *Experiment 3 (Traversal Mechanism over*

Uneven Terrain): This experiment evaluates the robot's decision-making when encountering obstacles, specifically whether it should roll over or step over them.

To keep the user's experience at the forefront of device iteration, experiments that involved user-in-the-loop testing (Experiments 1 & 2) were prioritized. Therefore, Experiment 3 was not tested as part of the MVP but should be evaluated as part of the final design.

Experiments 1 and 2 were performed with Raveena during a second round of user testing. For Experiment 1, her perceived safety and comfort were evaluated in several trials using three different paths. The independent variables, dependent variables, trial descriptions, and quantitative feedback of this experiment are detailed in Appendix AE. Overall, Raveena perceived the robot's operation as safe, with her highest perceived safety being in single-obstacle trials. Additionally, she never felt the need to stop or override the robot in any trial, as the robot's path changes were generally understandable and its motion was smooth (especially in the single-obstacle cases). However, performance dropped in the two-obstacle scenario in terms of safety and smoothness. Raveena also reported nervousness only during the two-obstacle scenario, suggesting that closely spaced obstacle navigation is a weakness of the device. This suggests that the robot's behavior became less interpretable in busier environments.

For Experiment 2, lengths of path segments were pre-determined using available routes near the student center. The testing route was selected to include multiple path options, allowing Raveena to choose whether the robot turned left or right at the end of the path segment. Nine trials were conducted in total, with three trials each using a 0.25m turning radius, 0.5m turning radius, and 0.75m turning radius. Feedback from Raveena indicated that the trials with a 0.5m turning radius felt the most comfortable and smooth. Raveena was given the option to select a right or left turn at random for each trial, and the device was able to detect her voice and execute the appropriate command in 8 of the 9 trials.

Robot-only Testing

Evaluation of individual MVP design criteria is detailed in Appendix AG. To test robot-only obstacle avoidance capabilities, a set of common sidewalk hazards were collected and categorized by sizes. These obstacles were used to evaluate how reliably the robot could detect hazards of varying sizes and navigate without collision. Appendix AH summarizes results from multiple testing rounds across 4 classes of obstacles. Trials were conducted by placing one or more obstacles in the robot's path and recording whether the robot successfully navigate around them without collision. Testing was repeated until each obstacle class had been evaluated across 10 total trials. Avoidance success is defined as the percentage of trials in which the robot successfully navigated around a given obstacle without making contact. From testing, it was observed that the robot struggled to detect thinner and smaller obstacles. This is likely due to the upwards pointing angle of the RealSense, occluding near range short obstacles as well as variability of depth information filtering. In future iterations, LiDAR integration is necessary for optimal object detection, and a downwards facing camera angle can be used to improve ground-level detection.

Final Design Solution

The final design solution is not meant to replace the user's agency in navigation, but instead acts similarly to a sighted guide and combines physical guidance, environmental context, and communication to make travel through unfamiliar urban environments feel safer and more intuitive. The final design works to fulfill the criteria defined in Table 3. The primary functionality of the final product that is not included in the MVP is destination-based guidance. The robot will incorporate GPS and grid-based path planning algorithms to guide the user through their desired outdoor route. While the MVP did not include full GPS-based navigation, it demonstrated the beginning of this behavior through guided assistance on preset routes and shared control through user-directed turns. This shows that the device can support the core interaction needed for future navigation. In addition, rudimentary evaluation of obstacle detection capabilities using the Realsense indicated that LIDAR integration and a second camera may need to be included in the final design to achieve the desired FOV. Specifically, the second camera should be angled towards the ground to detect obstacles and uneven terrain within 1m of the robot's front face.

A major consideration for the final design was developing a user interface that allows a level of user control, as a fully autonomous device can be unsafe and reduces the dignity of the BVIP using it. A detailed overview of the components of this interface can be found in Appendix AD. This also addresses the broader design goal of reducing cognitive load by allowing users to select their desired interaction style. In addition to physical criteria evaluated as part of the MVP, including robot speed and weight, the final product will include improvements to battery capacity, packaging, and weatherproofing that would allow it to support multiple round-trip journeys, reach up to 8 hours of use, and meet the intended IP54 water-resistance target. The final cost of the BOM of the device as presented in the fabrication package is \$4,040.63. Based on this, the market value of the device would be between \$6,000 - \$8,000. This allows the design to remain accessible while keeping advanced, user-focused functionality.

Risk Assessment, Safety and Liability

Risks associated with the final concept and corresponding mitigation strategies are listed in Appendix L.

Societal, Environmental, and Sustainability Considerations

A Societal Impact Assessment for SidewalkSidekick is in Appendix Z.

Conclusions and Future Work

Sidewalk Sidekick was developed to help BVIP navigate outdoor urban environments safely and confidently. The developed prototype integrates a wheeled quadruped mobility platform, onboard sensing, an open-ear headset, and a Bluetooth handle interface to support shared control between users and the device. The system demonstrated the ability to guide a user through controlled urban scenarios with both proprioceptive and auditory cues, respond to handle inputs, route around transient barriers, and provide basic surrounding contextualization. Prototype development, although far from a market-ready solution, provides a few main conclusions. First, robotic guidance devices can provide an intuitive assistive framework, where the device behaves as a sighted guide, rather than a fully autonomous system taking control away from users. Second, the combination of proprioceptive feedback through a rigid handle and auditory feedback enables greater awareness than existing mobility aids. Third, the use of a simple button-based interface for key functionality, and the use of speech recognition for more complex commands creates lower overhead for device operation. However, this early work also emphasized that safe robotic guidance depends heavily on the reliability of perception, localization, and effective communication. The current platform fuses sensor information for pose estimation, but lacks accuracy since odometry and IMU data contain noise.

Future work should focus on expanding device functionality, primarily in regions of sensing and developing more robot user control pathways. While the current system demonstrates feasibility for obstacle-aware guidance, global navigation, crosswalk handling, curb traversal, and more accurate localization are not developed. Additional sensors such as GPS, an IMU with integrated magnetometer, and more cameras should be explored to increase perception fidelity. The user interface should also be expanded to include outlined preferences in Appendix AD. The electromechanical design, while architecturally capable, should also be refined to support longer runtimes (>8 hrs), implement weather resistance, reduce pinch-points, and make the device easier to carry indoors. Through these improvements, SidewalkSidekick will be closer to a complete assistive device rather than its current prototyping state.

Team Member Contributions

Name	Contribution
Keerthi Padmanabhan	Industrial Design/Human Factors for Human-Robot Interface, SIA
Grant Espy	Electromechanical Design, Manufacturing, Fab Package, Final Design Solution, Conclusion & Future work
Sadhana Kumar	Communication Protocol, Stakeholder Feedback and Resulting Changes, User-in-the-loop testing, Exec Summary, Intro, Conclusion
Sahas Suri	Mechanical Architecture, Electrical Architecture, Software Architecture, MVP Feasibility Evaluation
Pranav Nair	Software Architecture, Realsense Feasibility Validation, Robot-only Testing, Obstacle Avoidance Testing
Vincent Griffo	Mechanical Architecture, Engineering Analysis, Fab Package

References

- [1] Lavric A, Beguni C, Zadobrischi E, Căilean AM, Avătămăniței SA. A Comprehensive Survey on Emerging Assistive Technologies for Visually Impaired Persons: Lighting the Path with Visible Light Communications and Artificial Intelligence Innovations. *Sensors* (Basel). 2024 Jul 25;24(15):4834. doi: 10.3390/s24154834. PMID: 39123881; PMCID: PMC11314945.
- [2] S. Nikanfar, A. Hebri, H. Ram Nambiappan, G. Nale, M. Siddiqua, F. Farhanipad, and F. Makedon, "A survey on assistive technologies for visually impaired individuals: Recent innovations, limitations, and future directions," in *Proc. 18th ACM Int. Conf. Pervasive Technol. Relat. Assistive Environ. (PETRA '25)*, New York, NY, USA: ACM, 2025, pp. 429–434, doi: 10.1145/3733155.3734895.
- [3] Y. Wei, N. Rocher, C. Gupta, M. H. Nguyen, R. Zimmermann, W. T. Ooi, C. Jouffrais, and S. Nanayakkara, "Human–robot interaction for blind and low vision people: A systematic literature review," in *Proc. CHI Conf. Human Factors Comput. Syst. (CHI '25)*, New York, NY, USA: ACM, 2025, Art. no. 276, pp. 1–19, doi: 10.1145/3706598.3713438.
- [4] Y. Gao, D. Wu, J. Song, X. Zhang, B. Hou, H. Liu, J. Liao, and L. Zhou, "A wearable obstacle avoidance device for visually impaired individuals with cross-modal learning," *Nature Communications*, 2025, doi: 10.1038/s41467-025-58085-x.
- [5] H. Hwang, H.-T. Jung, N. A. Giudice, J. Biswas, S. I. Lee, and D. Kim, "Towards robotic companions: Understanding handler–guide dog interactions for informed guide dog robot design," in *Proc. CHI Conf. Human Factors Comput. Syst. (CHI '24)*, Honolulu, HI, USA: ACM, 2024, Art. no. 596, pp. 1–20, doi: 10.1145/3613904.3642181.
- [6] L. Wang, Q. Chen, Y. Zhang, Z. Li, T. Yan, F. Wang, G. Zhou, and J. Gong, "Can quadruped navigation robots be used as guide dogs?" *arXiv preprint* arXiv:2210.08727, 2023. [Online]. Available: <https://arxiv.org/abs/2210.08727>
- [6] K. Seetharaman *et al.*, "Influence of the built environment on community mobility of people living with visual disabilities: a scoping review," *Urban, planning and transport research*, vol. 12, no. 1, Jan. 2024, doi: <https://doi.org/10.1080/21650020.2023.2296891>.
- [7] Cushley, Laura N., et al. "Navigating the Unseen City: Town Planners, Architects, Ophthalmic Professionals, and Charity Opinions on Navigating of the Built Environment with a Visual Impairment." *International Journal of Environmental Research and Public Health*, vol. 19, no. 12, 14 June 2022, p. 7299, <https://doi.org/10.3390/ijerph19127299>.
- [8] F. El-Taher, L. Miralles-Pechuán, J. Courtney, K. Millar, C. Smith, and S. McKeever, "A survey on outdoor navigation applications for people with visual impairments," *IEEE Access*, vol. 11, pp. 14647–14666, 2023, doi: 10.1109/ACCESS.2023.3244073.
- [9] "FAQs - Guide Dogs of America." *Guide Dogs of America*, 19 May 2025, www.guidedogsofamerica.org/frequently-asked-questions/.
- [10] "Assistive Technologies for the Visually Impaired Market," *Yahoo Finance*, May 29, 2025. [Online]. Available: <https://finance.yahoo.com/news/assistive-technologies-visually-impaired-market-081400737.html>

- [11] Dutta, Tushmi. "Blind Assistive Technology Market." *Nextmsc.com*, Next Move Strategy Consulting, 9 Dec. 2025, www.nextmsc.com/report/blind-assistive-technology-market-3858. Accessed 10 Feb. 2026.
- [12] "Meet Guide Dogs for the Blind." *Guide Dogs for the Blind*, www.guidedogs.com/about-guide-dogs-for-the-blind/meet-guide-dogs-for-the-blind.
- [13] P. Mo, G. Lyu, H. Fan, Z. Liu, A. Guin and R. Guensler, "Sidewalk Ada Design and Anomaly Condition Assessment Using Gopro Vibration Data," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 3751-3758, doi: 10.1109/ITSC58415.2024.10920184.
- [14] "1910.25 - Stairways. | Occupational Safety and Health Administration," [www.osha.gov. https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.25](https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.25)
- [15] Federal Highway Administration, University Course on Bicycle and Pedestrian Transportation: Lesson 9—Walkways, Sidewalks, and Public Spaces. Washington, DC, USA: U.S. Department of Transportation, Jul. 2006.
- [16] A. Daryabor, M. Arazpour, G. Sharifi, M. Ahmadi Bani, A. Aboutorabi, and N. Golchin, "Gait and energy consumption in adolescent idiopathic scoliosis: A literature review," *Ann. Phys. Rehabil. Med.*, vol. 60, no. 2, pp. 107–116, 2017, doi: 10.1016/j.rehab.2016.10.008.
- [17] OSHA, "Slips, Trips, and Falls: Preventing Workplace Trip Hazards," [www.osha.com](https://www.osha.com/blog/slips-trips-falls-prevention), Feb. 22, 2023. <https://www.osha.com/blog/slips-trips-falls-prevention>
- [18] T. P. McAlexander, R. R. Gershon, and R. L. Neitzel, "Street-level noise in an urban setting: Assessment and contribution to personal exposure," *Environ. Health*, vol. 14, p. 18, Feb. 2015, doi: 10.1186/s12940-015-0006-y.
- [19] J. M. García and F. G. Duarte, "Mobile rolling robots designed to overcome obstacles: A review," *Forces in Mechanics*, vol. 16, Aug. 2024, Art. no. 100283. doi: 10.1016/j.finmec.2024.100283.
- [20] J. Zhang, J. Liu, H. Zong, P. Ji, and B. Xu, "Bridging the gap to bionic motion: Challenges in legged robot limb unit design, modeling, and control," *Cyborg and Bionic Systems*, vol. 6, Art. no. 0365, Aug. 2025, doi: 10.34133/cbsystems.0365.
- [21] Y. Xue, W. Wang, M. Duan, N. Jiang, S. Zhang, and X. Xiao, "Rolling vs. swing: A strategy for enhancing locomotion speed and stability in legged robots," *Biomimetics*, vol. 10, no. 7, Art. no. 435, Jul. 2025, doi: 10.3390/biomimetics10070435.
- [22] J. E. McKenzie, *Design of Robotic Quadruped Legs*, M.S. thesis, Dept. Mech. Eng., Massachusetts Institute of Technology, Cambridge, MA, USA, Feb. 2012.
- [23] H. Dreyfuss, *The Measure of Man: Human Factors in Design*. New York, NY, USA: Whitney Library of Design, 1967. Accessed: Mar. 20, 2026. [Online]. Available: https://ia801906.us.archive.org/34/items/TheMeasureOfManDreyfuss/The%20Measure%20of%20Man%20%28Dreyfuss%29_text.pdf

[24] C. C. Gordon et al., "2012 Anthropometric Survey of U.S. Army Personnel: Methods and Summary Statistics," Natick Soldier Research, Development and Engineering Center, Natick, MA, USA, Tech. Rep. NATICK/TR-15/007, 2015. [Online]. Available: DTIC ADA611869

[25] J. Tamparong, "How Polyurethane Compares to Other Shock Absorbing Foams," *General Plastics*, Sep. 07, 2025. <https://www.generalplastics.com/blog/how-polyurethane-compares-to-other-shock-absorbing-foams>

Appendices

Appendix A: Comparison of relevant predicate technologies

Name	Image	Premise	Core Capability	Primary Drawback
<i>Traditional White Cane</i>		Passive tactile aid	Ground-level obstacle detection with human-in-the-loop control	No proprioceptive guidance; slower navigation due to required human processing
<i>Guide Dog</i>		Biological mobility system	Obstacle avoidance and trained proprioceptive route intelligence without continuous human-in-the-loop control	High cost, limited scalability, requires retraining for new routes
<i>Maptic™</i>		Wearable haptic system	Upper-body obstacle detection with navigation guidance	No ground-level obstacle detection; no proprioceptive guidance
<i>Microsoft Soundscape</i>		Audio navigation application	Spatial audio navigation during path traversal	No obstacle detection; no proprioceptive guidance
<i>WeWalk</i>		Smart cane system	Chest-level obstacle detection with integrated app-based navigation	No ground-level obstacle detection; no proprioceptive guidance
<i>Glidance (Glide™)</i>		Robotic mobility aid	Active obstacle avoidance with proprioceptive guidance	Limited terrain versatility and constrained user feedback; cannot traverse transient barriers
<i>Robot Guide Dog (R&D)</i>		Legged robotic platform	Autonomous proprioceptive navigation and obstacle avoidance over uneven terrain	High cost, low design maturity, reduced user trust
<i>GUIDi™</i>		Wearable AI navigation system	Computer vision-based obstacle avoidance with haptic feedback	Limited physical obstacle interaction; no proprioceptive guidance

Appendix B: Stakeholder summary and key findings

Throughout the design process, stakeholders have been updated with the criteria table and design concepts. Feedback from these individuals has been instrumental in modifying the design so that it fits the target community.

With the stakeholders presented in Table 1, it was imperative to employ qualitative interviews, focus groups, site visits, and expert consultations to ensure a comprehensive understanding of BVIP needs. Across these methods, customer discovery style frameworks were used to minimize bias and open-ended prompts included the following:

- “Walk us through your typical commute to work”
- “Tell us about the last time you felt unsafe or frustrated while traveling.”
- “What technologies have you used? What do you like and dislike about them?”

Open discussions with our stakeholders provided valuable insight into everyday challenges faced by visually impaired individuals in city environments as well as bottlenecks with existing assistive technologies.

Site visits to Tools for Life, an organization comprised of assistive technology specialists, provided insights into current assistive devices and their popularity among the BVIP community. Additionally, conversations about fitting new tools into existing ecosystems involving white canes and smartphone apps provided valuable perspectives regarding the importance of aesthetics and stigma minimization with assistive technology device design. To supplement the testimonials from our customer discovery efforts, literature was referenced to that included more testimonials and insights from the BVIP community.

Key Finding 1: Urban outdoor navigation is the primary pain point. Among stakeholders, a predominant challenge was navigating urban outdoor environments. 4 of the 5 stakeholders we interviewed mentioned frequent encounters with urban obstacles like scooters, trash bags, parked cars, low hanging branches, and tree roots and hazardous terrain like broken or uneven pavements, and puddles as major hindrances to urban navigation. These findings were also consistent among published reports of unsafe sidewalks and transient obstacles. In fact, one journal cited 43 peer reviewed studies in the past 22 years from North America, Europe, and Oceania to understand how built environment features affect mobility among the visually impaired community [6]. The same obstacles mentioned by our stakeholders were addressed in this comprehensive study.

Key Finding 2: High injury risk persists despite exiting mobility aids. These barriers that were identified among the population have been shown to increase the risk of falls, disorientation, and fear [7]. 63% of respondents indicated they had suffered an injury on at least one previous occasion when navigating outdoors, likely due to the limited ground obstacle detection scope offered by white canes [8]. From our own stakeholder interviews, however, identification of these barriers represented only a fraction of the urban navigation challenge.

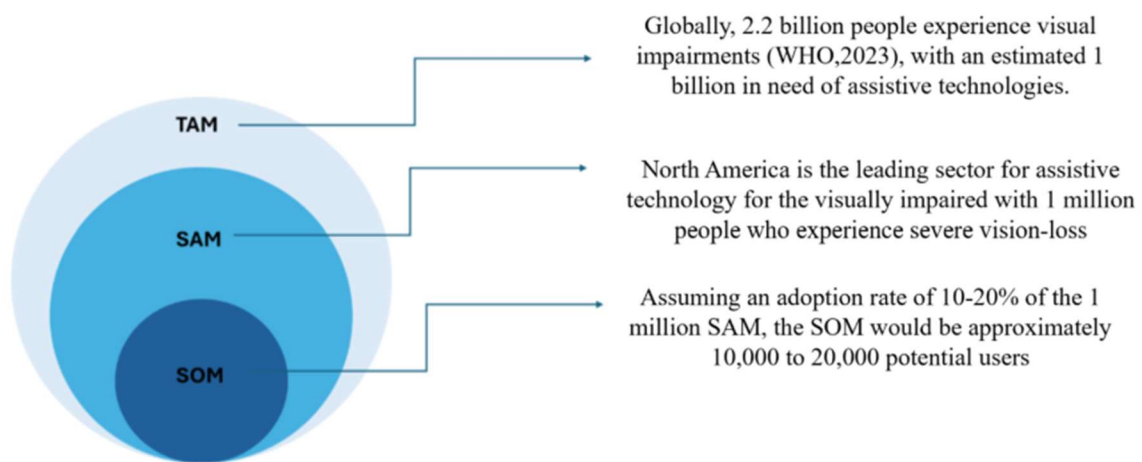
Key Finding 3: Technology adoption is constrained by stigma and cognitive load. A bulk of the BVIP urban navigation challenge refers to the stigma surrounding assistive technology usage. While participants expressed interest in assistive technology, adoption depended largely on the device’s social conspicuity and appearance. It was also imperative that new assistive technologies relied on intuitive and non - overwhelming feedback. Multi-sensory feedback was valued but only when carefully balanced to avoid cognitive overload. A central insight from assistive technology specialists was the importance of designing solutions that minimize visible, bulky, or awkward interactions to reduce social stigma.

Key Finding 4: Traditional aids are essential but have their own drawbacks. White canes and guide dogs are fundamental mobility tools for the BVIP community. However, our stakeholders reported limited ground level detection for transient barriers such as scooters and above ground obstacles such as tree branches. Guide dogs, while invaluable to the BVIP community, are costly with the average cost to breed,

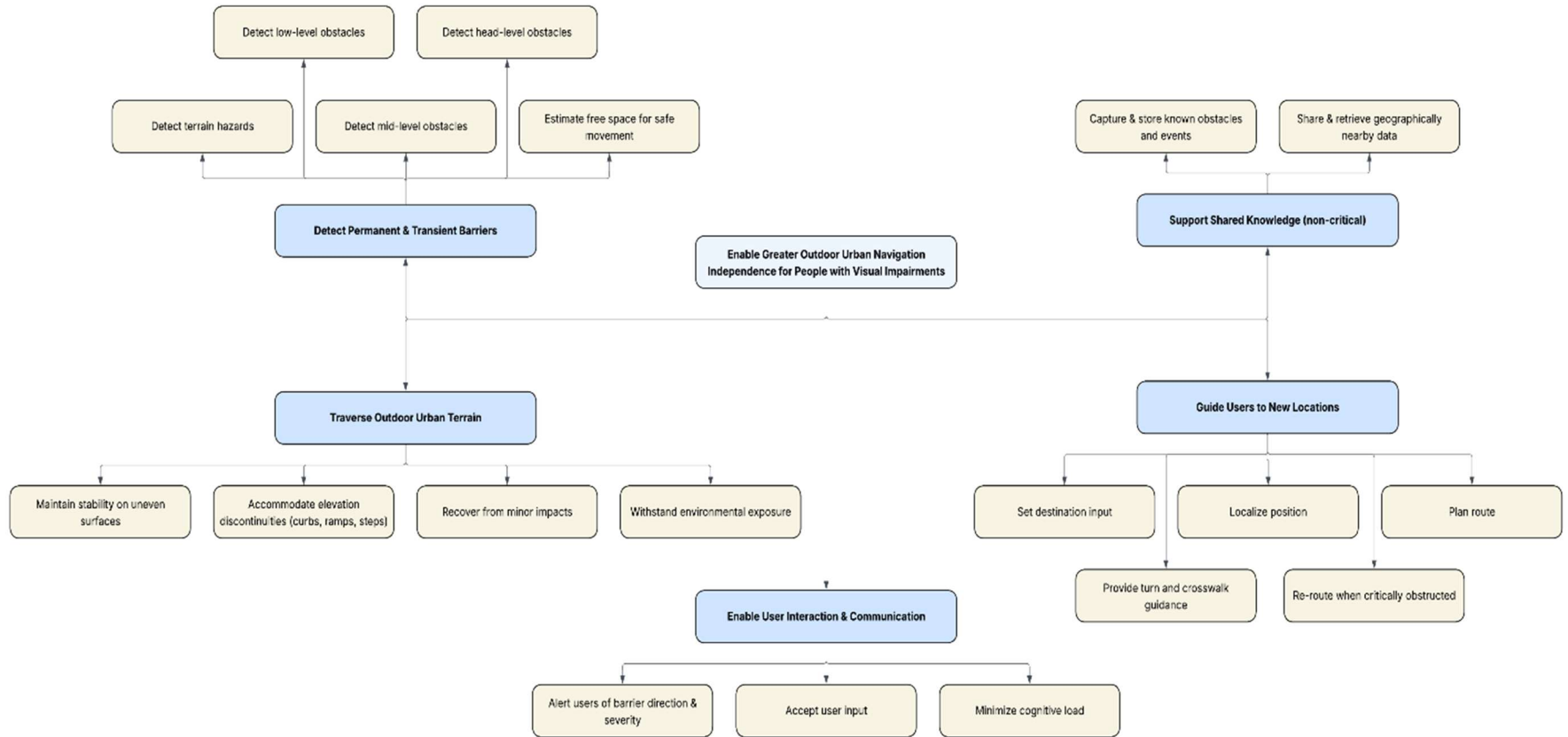
raise, and train a service dog being \$60,000 and wait times for service dogs can be up to three years [9]. Additionally, one stakeholder mentioned the added burden of maintenance and responsibility associated with a guide animal. Building on these findings, a mobility device that strengthens navigation and object detection would offer visually impaired individuals with greater confidence and security in dynamic urban environments.

Appendix C: Market Analysis & Growth

Broader blind assistive technology such as smart canes, AI-powered wearables, and navigation apps was valued around \$7.13 billion in 2025, with North America leading this sector with 36.07% of the market share [10][11]. The market for visually assisted technologies is expected to grow at a compound annual growth rate of 12.67% over the next four years, with major players like Apple, Google, and Plantronics operating in this space. In contrast, access to traditional mobility aids remains limited. In the United States and Canada, there is expected to be only 10,000 to 20,000 active service dogs, despite, 1 million Americans experiencing severe vision loss [12]. As a result, the demand for service dogs is unable to meet the need for assistive guidance technologies. If just 10-20% of the 1 million Americans experiencing severe vision loss were to adopt the new assistive technology that we seek to develop, the estimated service addressable market (SOM) would be 100,000 to 200,000 users in the US alone. A market analysis is shown below.



Appendix D: Function tree of critical design features and their subfunctions



Appendix E: Description of core device functionality and associated evaluation metrics

Traverse Outdoor Urban Terrain serves to ensure the device can move alongside end-users across sidewalks, streets, and crosswalks. Key metrics used to evaluate performance in urban terrain include maximum traversable vertical height, minimum turning radius, and minimum tipping force. Wearables are not evaluated on the same criteria, since they travel with the user regardless of terrain.

Detect Permanent and Transient Barriers ensures that the platform can identify a range of obstacles known to limit mobility for BVIP. Minimizing latency provides enough lead time for users to react to alerts and redirect themselves, preventing injury. Concepts for detection will be evaluated according to sensing range, latency, minimum obstacle size for effective detection, and overall accuracy.

Enable User Interaction and Communication encompasses providing an intuitive interface for accepting inputs and relaying information. Important criteria related to user interface include alert interpretability, command response time, and user training time.

Guide Users to New Locations is the last core function of the device, since mobility for BVIP is greatly reduced due to lack of spatial context or existing memory. Integrating journey routing capabilities with a platform centered around hazard avoidance allows for increased confidence during independent travel as compared to existing applications such as Google Maps. Effectiveness can be measured via route quality metrics, localization accuracy, and re-routing time in the presence of impassable hazards.

Support Shared Knowledge is a non-critical function that would be easy to integrate in later stages of product development. Such a feature would enable distribution and access to relevant information to improve route planning ahead of time and avoid severe hazards.

Appendix F: Engineering specification table for fully featured product (not MVP)

<i>Functional Category</i>	<i>Demand vs Wish</i>	<i>Technical Requirement</i>	<i>Rationale/Source</i>	<i>Priority</i>
<i>Traverse Outdoor Urban Terrain</i>	D	Maintain stable operation on terrain with local surface height variations up to 38 mm (1.5 in)	Surface roughness beyond this threshold is considered hazardous for individuals with disabilities – should be avoided via rerouting [13]	High
	D	Able to traverse vertical discontinuities between 100-240 mm (4-9.5 in)	OSHA minimum curb height and maximum stair riser height [14]	High
	D	Achieve minimum turning radius of ≤ 0.6 m	Enables 90° turns while remaining centered in the smallest width sidewalk of 1.2 m [15]	High
	W	Support operating speeds up to ≥ 1.24 m/s	Mean adult walking speed reported in pedestrian mobility study [16]	Medium
	D	Ingress Protection (IP) rating IP54	Required for outdoor urban operation [IEC 60529]	High
<i>Detect Permanent & Transient Barriers</i>	D	Identify terrain height variations or gaps ≥ 6.35 mm ($\frac{1}{4}$ ")	OSHA tripping hazard minimum height [17]	High
	D	Detect obstacles within 0-5 m range	Provides ~4-5 seconds of lead time at walking speeds	High
	D	Detect obstacles from ground level to 2 m (6'6")	Accommodates head-level obstacles for 99% male height	High
	D	Perception latency of ≤ 250 ms	Maintains ability for average user to respond to obstacles as close as 50 cm	Medium
<i>Enable User Control & Interaction</i>	D	Provide configurable auditory and tactile feedback	Driven by user preferences	High
	D	Voice command recognition with up to 73.4 dBA background noise	Study of mean urban street noise levels [18]	Medium
<i>Guide & Navigate Users to New Destinations</i>	D	Average absolute localization accuracy of 5 m	Derived from current GPS technology accuracy	High
	D	Relative localization accuracy of ≤ 30 cm	Enables sidewalk positioning and obstacle avoidance in at least 4 zones	High
<i>Support Community Data Sharing</i>	W	Capable of storing and retrieving hazard information from a geotagged database	Shared experience and first-hand knowledge pool between users	Low
	W	Support capture and storage of video (≥ 3 min,	Occasional need for later reference or sighted individual guidance	Low

		~1.5 GB) for emergency situations		
<i>Packaging</i>	D	Device volume $\leq 350 \times 350 \times 350$ mm	Allows for safe use on walkways, enabling pedestrian passing	High
	D	Total weight ≤ 15 kg	NIOSH lifting guidelines and safety factor of 1.25	High
<i>Cost</i>	W	Total BOM $\leq \$1000$	Maintains ability to price as a low-cost assistive technology	Low

Appendix G: Engineering design standards applicable to SidewalkSidekick

There are several engineering design standards related to robotic design, haptic interaction, and environmental context that are relevant to this project. The applicability of each standard is classified as:

- **Direct:** the standard applies directly to the robot's intended use and informs core design requirements.
- **Partial:** there are relevant aspects of the standard (e.g. mobility or environment), but it does not fully capture the robot's assistive purpose.
- **Indirect:** the standard does not formally apply to the robot but can be used as guidance to inform safe design choices.

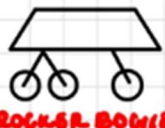
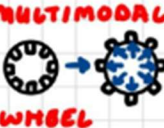
Applicable engineering standards are summarized in the table below. In addition to formal ISO/ANSI/ASTM standards, design decisions will also be informed by guidance from user organizations such as the National Federation of the Blind. This primarily addresses ergonomic factors such as device height and posture to reduce wrist and shoulder strain.

Table 4. Engineering design standards and corresponding design implications

	<i>Category</i>	<i>Standard</i>	<i>Relevance</i>	<i>Design Implications</i>	<i>Applicability</i>
1	Mechanical	ISO 13482:2014	For robots operating in close physical proximity to humans in non-industrial environments	Limits on speed and force; requires safe failure modes	Direct
2	Mechanical	ISO 22166-1:2021	Defines standardized terminology and classification for mobile robots and autonomy	Used to consistently describe system architecture and guidance behavior	Direct
3	Environmental	IEC 60529 (Target IP54)	Defines ingress protection ratings for dust and water exposure	Enclosure sealing and packaging to support outdoor operation	Direct
4	Human Factors	ISO 9241-910:2011	Framework for designing tactile & haptic interactions	Guidance on haptic cue design for clarity and comfort	Direct
5	Human Factors	ISO 9241-210:2019	Establishes human-centered design process for interactive systems	Iterative user research and testing; refinement based on user feedback	Direct
6	Safety	OSHA 29 CFR 1910 Subpart D	Slip, trip, and fall hazards for general walking-working surfaces	Identification of uneven terrain at minimum $\pm 1/4$ inches from baseline	Direct
7	Mechanical	ANSI 3100	Safety guidance for self-propelled mobile robots operating in shared spaces	Limits on speed; guidance on obstacle avoidance and emergency stop behavior	Partial
8	Environmental	ASTM F3218-19	Environmental and terrain considerations for outdoor mobile robots	Guidance on testing for outdoor terrain, weather, and surface variability	Partial

9	Safety	IEC 60601-1	Guidance for electrical and thermal safety in body-worn devices	Implement conservative electrical and thermal design choices	Indirect
10	Software	ASTM F3265-17	Test method for grid-based obstacle measurement via video	Testing implementation to guide successful and safe navigation around obstacles	Partial
11	Mechanical	ISO 22166-1:2021	Open modular design framework for service robots	Implementation of modular design principles in design	Indirect

Appendix H: Morphological chart for design concept ideation

Navigating Urban Terrain	 ROCKER BOWIE	 QUADRUPED	 MULTIMODAL WHEEL	 MODULAR CANE	 WEARABLE
Detection of Transient Barriers	 SONAR	 LIDAR	 CAMERA (CV)	 ACOUSTIC SENSOR	 IMU/ENCODER
Easy user Interaction	 BUTTONS	 HAPTIC FEEDBACK	 APPS	 AUDIO	 VOICE CONTROL
Guidance	 WIRELESS MAP	 PATH FOLLOWING			
Community	 HAZARD DATABASE	 SOCIAL MEDIA APP			

Appendix I: Solution architecture concepts

<i>Concept Name</i>	<i>Advantages</i>	<i>Disadvantages</i>
<i>Wheeled Rover with Suspension</i>	✓ Affordable relative to other robotic architectures ✓ Faster straight-line movement ✓ Rocker linkage and double wishbone suspension enable curb climbing	✗ Physically and socially obtrusive ✗ Wheel weight reduces ease of transport ✗ Unable to shift body to change sensor angle
<i>Quadruped Robot</i>	✓ Reduced social stigma due to familiar dog-like form factor ✓ More dynamic and maneuverable than wheeled concepts ✓ Increased flexibility of sensor positioning	✗ High mechanical and control complexity ✗ Slower than wheeled alternative ✗ Increased power draw ✗ Higher cost due to number of actuators and associated maintenance
<i>Transformable Wheel Rover</i>	✓ Easier traversal over urban terrain ✓ Maintains speed over obstacles ✓ Proprioceptive feedback informs user of uneven ground	✗ Physically and socially obtrusive ✗ Increased power draw due to weight and additional actuators ✗ Increased mechanical and control complexity ✗ Increased cost
<i>Modular Cane with Acoustic Sensor</i>	✓ Intuitive design builds on already familiar mobility aid ✓ User stays in control during use ✓ Works in obscured environments such as low-light, fog, rain ✓ Acoustic sensor can detect farther barriers than traditional cane	✗ Acoustic sensing is sensitive to echoes and urban noise interference ✗ Cannot guide users to new destinations ✗ Difficult to switch tips during use if terrain changes
<i>Wearable Navigation Belt</i>	✓ Non-obstructive ✓ Increased user comfort due to adjustable strap and weight ✓ Minimal feedback to not overwhelm user	✗ Clothing could obscure camera ✗ Field of view may not extend low enough to detect ground-level obstacles and tripping hazards ✗ Wearable device may clash with user's stylistic preferences

Appendix J: Scoring guidelines used for concept down selection

Score	Obstacle Detection Capability	Terrain Mobility & Adaptability	Guidance & Route Execution	Ease of User Interaction	Form Factor & Physical Obtrusiveness	Robustness, Endurance, and System Complexity	Total Cost
5 (Fully meets requirements)	Detects hazards at all levels (terrain, ground, mid, and head-level) up to 5 m ahead	Usable in standard urban terrain (cracks, curbs, uneven pavement) while maintaining stability	Provides continuous guidance around obstacles and along new routes; capable of rerouting	Minimal learning curve and cognitive load; supports multimodal notifications based on preference	Compact, lightweight, socially unobtrusive, minimal physical strain during use	Designed for outdoor use with minimal maintenance and system complexity	Very affordable $\leq$ \$500
3 (Partially meets with limitations)	Detects some hazards up to 5 m ahead	Usable over some terrain but with limited mobility or robustness	Takes users from point-to-point with limited obstacle avoidance	Requires moderate memorization and cognitive load; multimodal notification	Noticable size and weight, but minimal physical strain during use	Designed for outdoor use with high system complexity	Priced similarly to comparable assistive technology $\sim$ \$1500
1 (Does not meet core requirement)	Inconsistent detection and limited sensing range	Cannot be utilized over standard urban terrain variations	Does not guide users to new destinations	Complex interface with no preference settings	Highly obtrusive, heavy, and causes physical strain	Not suitable for regular outdoor use, highly complex, or prone to damage	More expensive than comparable technology $\geq$ \$2000

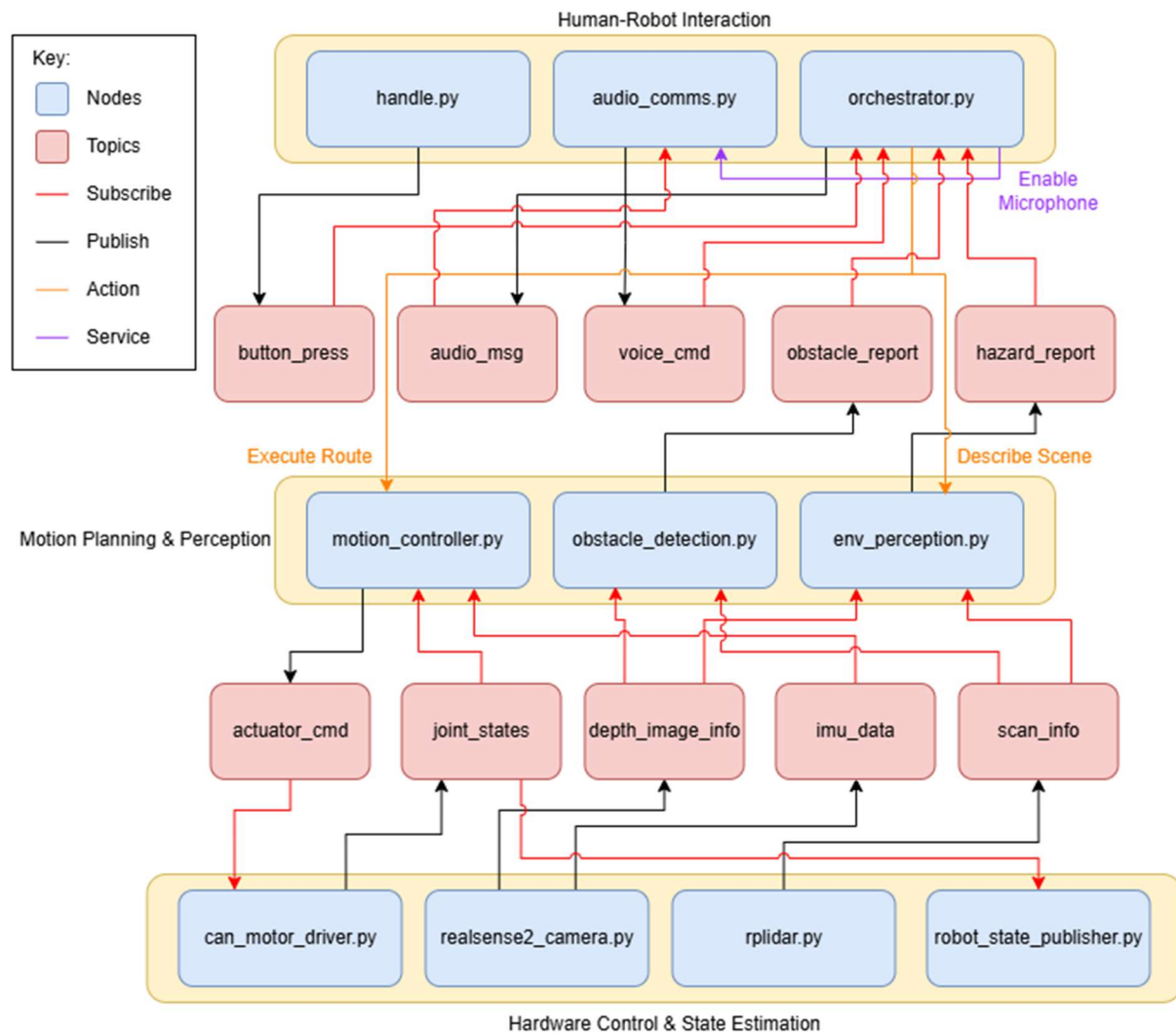
Appendix K: Weighted downselection matrix showing quadruped architecture is most suitable for meeting customer needs

	Criteria 1	Criteria 2	Criteria 3	Criteria 4	Criteria 5	Criteria 6	Criteria 7	
Criteria Description	Can the system reliably detect hazards across all levels and quickly enough to keep end-user safe?	Can the device be used across varying outdoor urban terrain?	How well does the system guide people with visual impairments during commute?	How easy is it to interpret and influence the systems behavior?	How unobtrusive is the platform?	Will the device perform as expected and survive frequent use? Is it easy to maintain and minimally complex?	How much is the system expected to cost, contextualized by price-points of other assistive technology?	
	Obstacle Detection Capability	Terrain Mobility & Adaptability	Guidance & Route Execution	Ease of User Interaction	Form Factor & Physical Obtrusiveness	Robustness, Endurance, and System Complexity	Total Cost	WEIGHTED SCORE
Weight	5	4	4	5	3	3	1	25
	20.00%	16.00%	16.00%	20.00%	12.00%	12.00%	4.00%	100.00%
Solution Concepts	Criteria 1 SCORES	Criteria 2 SCORES	Criteria 3 SCORES	Criteria 4 SCORES	Criteria 5 SCORES	Criteria 5 SCORES	Criteria 6 SCORES	
Wheeled Robot with Rocker/Double Wishbone Suspension	5	3	4	2	1	5	3	3.36
Quadruped Robot	5	5	5	5	3	3	2	4.4
Transformable Wheel Rover	5	4	4	2	2	2	3	3.28
Modular Cane with Acoustic Sensing	1	3	1	4	4	2	5	2.56
Wearable Haptic Navigation Belt	2	5	3	3	4	3	4	3.28

Appendix L: Risks and potential mitigation strategies associated with wheeled quadruped architecture

<i>Risk</i>	<i>Causes</i>	<i>Potential Mitigation Strategies</i>
<i>Audio Communication Failure</i>	Bluetooth disconnects, headphones fall out, ambient noise masks audio input	▪ Incorporate multiple user interaction channels, e.g. bone conduction audio headset, haptic belt, handle vibration, force cues via handle ▪ Physical override button on handle ▪ Auto-stop if communication is lost for > 2 minutes
<i>Loss of Control</i>	Incorrect path planning, wheel slip causes odometry confusion	▪ Combine inputs from multiple sensors to increase spatial and environmental data collection accuracy ▪ Ambulatory speed limits in shared spaces ▪ MPC control algorithm to avoid reaching joint and torque limits ▪ Safe stop behavior when path uncertainty crosses a specified threshold
<i>Mechanical Instability</i>	Wheel unknowingly encounters curb, sudden lateral force from user	▪ Design around a low center of mass ▪ Force sensing in handle ▪ Fall detection and subsequent system shutdown
<i>Physical Injury to User</i>	Control algorithm causes sudden acceleration, user trips while following robot	▪ Speed capped to ambulatory pace (or customizable pace setting) ▪ MPC control imposes torque limits ▪ Jerk-limited motion planning
<i>Power Failure</i>	Battery dies mid-route	▪ Audible low battery warning ▪ Conversion to manual mode while ensuring robot weight does not cause joint strain ▪ Portable charging option
<i>Robot Collides with Pedestrians</i>	Crowded urban environment	▪ Detect dense crowds and reduce speed accordingly ▪ Audible signaling or indicator light to show planned motion ▪ Design or color scheme enforces visibility of the device
<i>Cognitive Overload for User</i>	Overwhelming or confusing feedback and navigation	▪ Minimal and consistent vocabulary and cues ▪ Option to start in “training mode” before allowing the device to guide the user autonomously

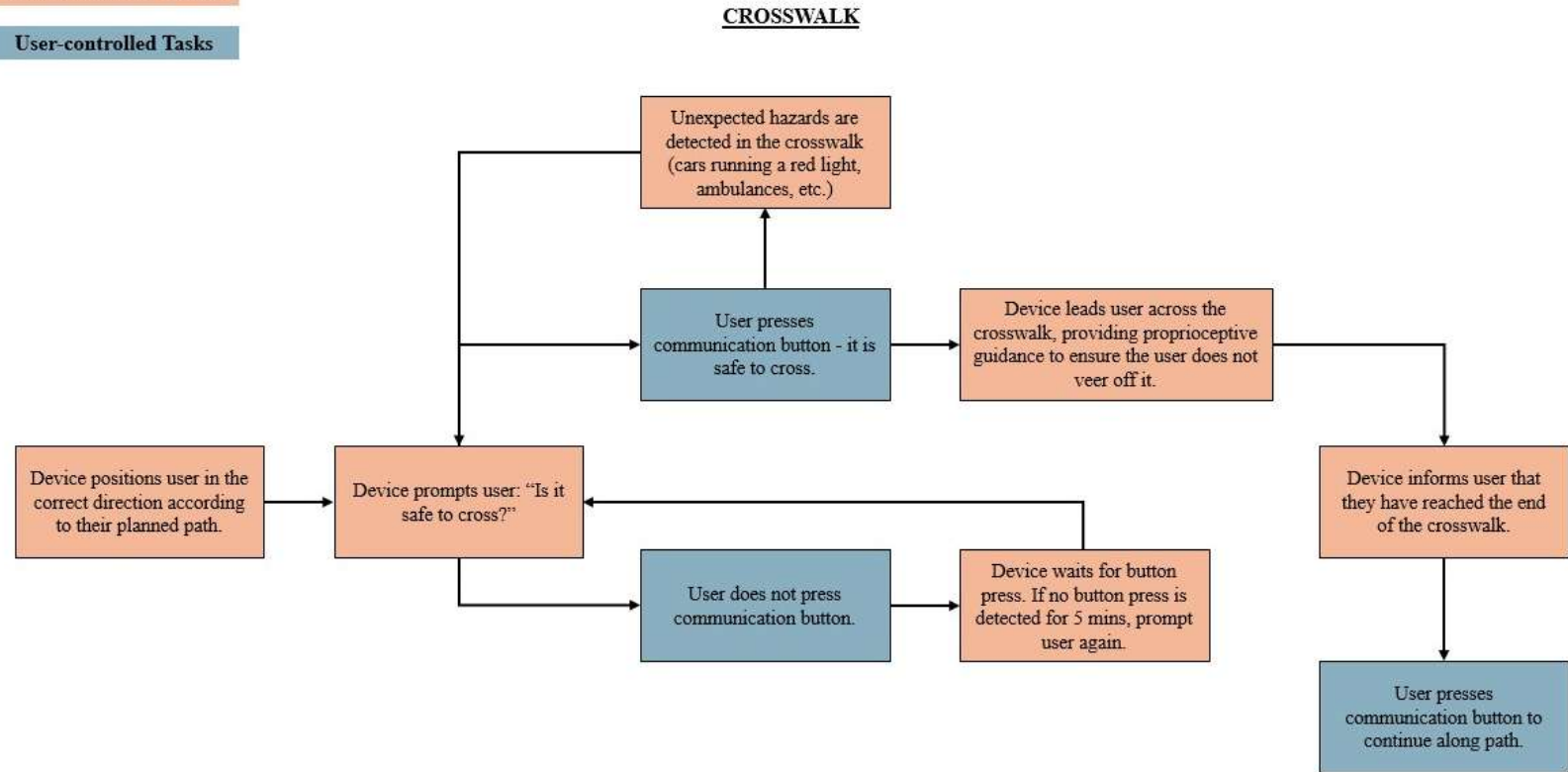
Appendix M: Preliminary node graph based on ROS 2 framework for MVP feasibility demonstration



Appendix N: User control flowchart for crosswalks

Device-controlled Tasks

User-controlled Tasks

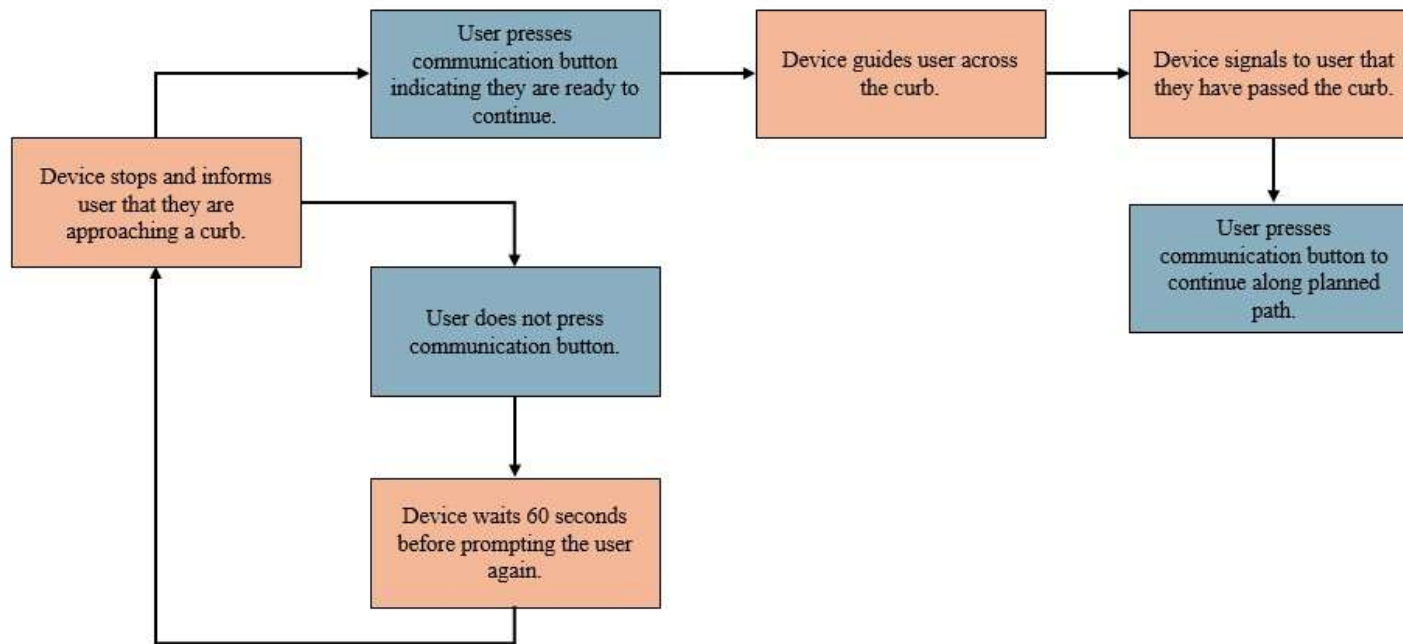


Appendix O: User control flowchart for curbs

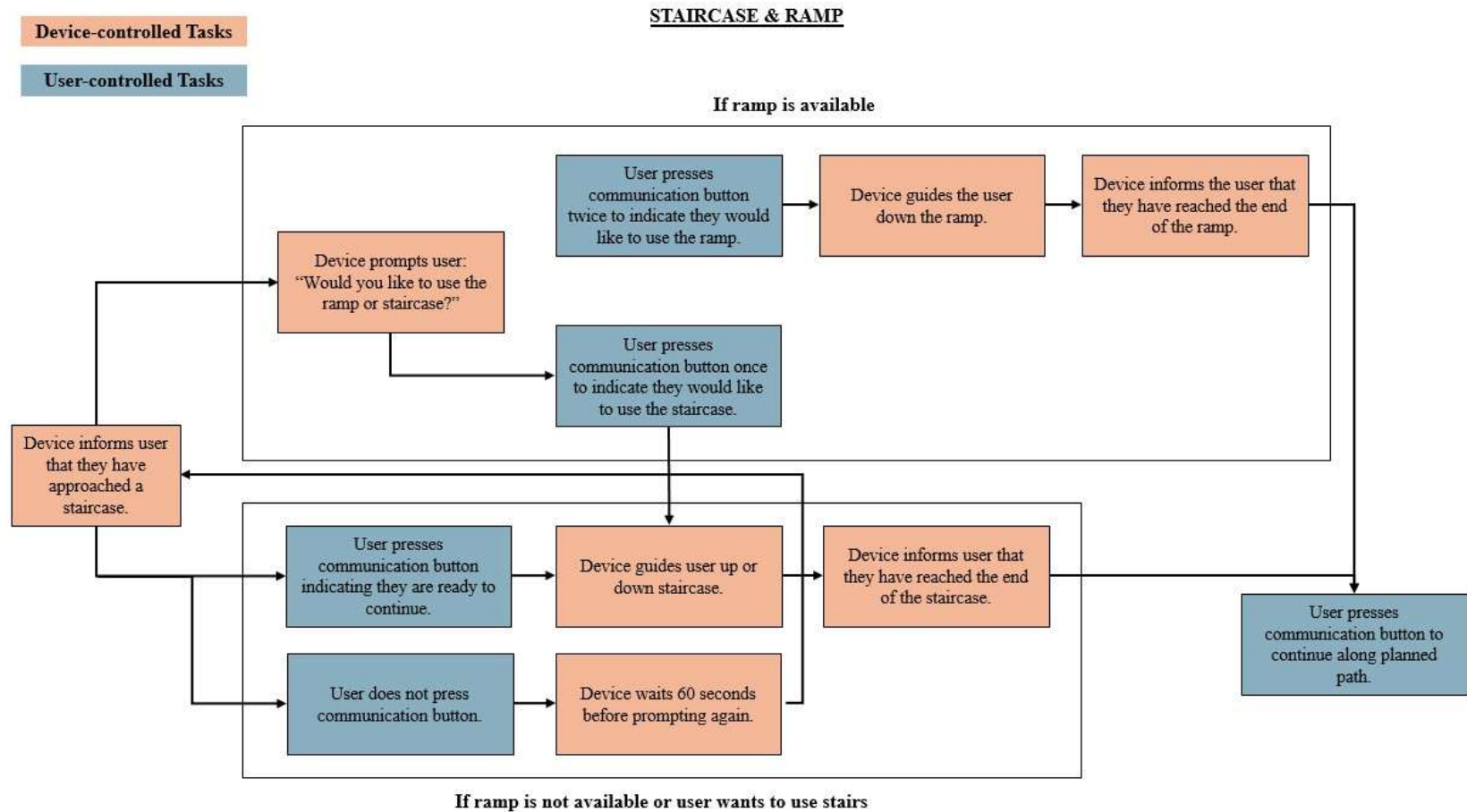
Device-controlled Tasks

User-controlled Tasks

CURB



Appendix P: User control flowchart for staircases and ramps



Appendix Q: Potential questions for stakeholder evaluation of handle prototypes

Effective Communication & Responsiveness

- How quickly does the handle respond when you press the buttons? (1-5, 5 = immediate, 1= very slow)
- How well do these buttons signal robot intent? (1-5, 5= very well, 1 = poor)

Secure Grip and Posture

- Rate your grip security with this handle (1-5, 5 = very secure, 1 = not secure)
- How do your thumb and wrist feel after 2 minutes?

Tactile & Visual Button Identifiability

- How easy is it for you to find all the buttons with touch alone (1-5, 5 = very easy, 1 = very difficult)

Familiarity & Low Learning Curve

- What does this remind you of?
- How long does it take you to familiarize yourself with the device?

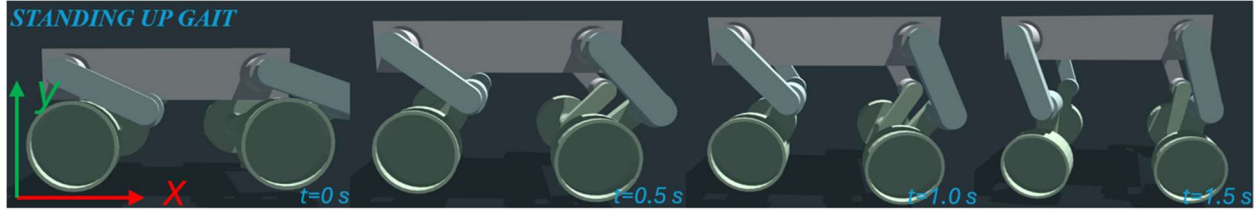
Autonomy vs Guided Navigation Balance

- Describe how you feel? How is the balance between being led vs controlling your path?
- How easy is it to override robot guidance (1-5, 5 = very easy, 1 = very difficult)

Appendix R: Feedback received from stakeholders and resulting design changes

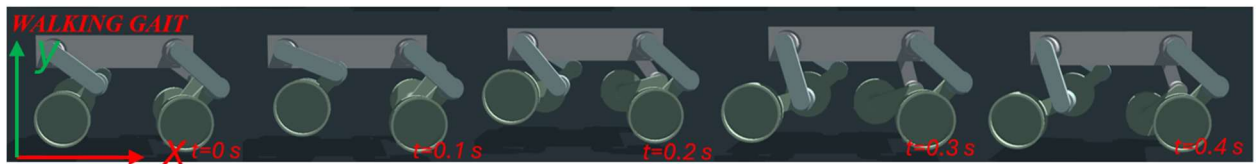
<i>Stakeholder</i>	<i>Feedback</i>	<i>Resulting Changes</i>
Jennifer Ottowitz (BVI, Research Specialist)	“I recommend using both visual contrast and tactual distinction if possible... the buttons should be a contrasting color from the surface of the device... many people who are visually impaired have some amount of vision and would find this helpful”	Button designs feature visual contrast and tactile distinction to cater to all levels of visual impairment
	“Any type of handle or portion held by the person should fit comfortably in the hand even when wearing gloves”	ANSUR database is referenced for handle design
	“I’m curious what the feedback will say about unavoidable obstacles... if ‘unavoidable’ is used, the user may think this is unsurmountable”	Wording in communication protocol changed from ‘Unavoidable Obstacles’ to ‘Obstacles above Waist Level’ to preserve user dignity
	“Will the device line up the person with the curb cut and give feedback if the person veers while crossing?”	User control workflow for crosswalks (Appendix N) updated to include that the robot will line the user up with their planned path and provide proprioceptive guidance to stay on crosswalk
	“It would be nice for the user to have a choice to find the ramp if one is available as sometimes a ramp may not be desired because it is a longer distance away”	User control workflow for staircases and ramps (Appendix P) updated to include multiple interaction pathways depending on the availability of a ramp
	“Will the device inform of a curb as well as any stairs?”	Separate user control workflow for curbs was created (Appendix O)
Keely Zeitlin (O&M Specialist at CVI)	Symbols for buttons are preferred over Braille text descriptors	Shape of buttons designed to inherently describe their function
	Familiarity of the handle is an important consideration from a training perspective	Switch from retractable leash to handle interfaces that are preferred by guide dog users
	Hard to reorient hand and buttons if handle is dropped when using a retractable leash	Switch from retractable leash to designs with 1 passive DOF
Danny Housley (BVI, Assistive Technology Case Manager)	“Tension is important when it comes to starting/ stopping; this is why many people will use the handle/harness for service dogs... you get more immediate feedback on when to stop rather than waiting on the leash to slacken”	Switch from retractable leash to more rigid and centered handle interfaces that allow for easier interpretation of proprioceptive guidance
	“I like the level of communication control, so that detours and independent route planning can occur”	N/A
	“I like the idea of this being a navigation aid and not an unwavering guide”	N/A
	“Wheels rather than uncanny valley-esque walking will help with not garnering undue attention”	N/A
	“Have you considered increasing visibility for this device? Reflective highlights, possibly lights for when traveling at night...”	Further discussions will be held on the color of the device (white and red) and reflectivity of the material will be emphasized for final design

Appendix S: Showcase of different gait paths of interest for Sidewalk Sidekick



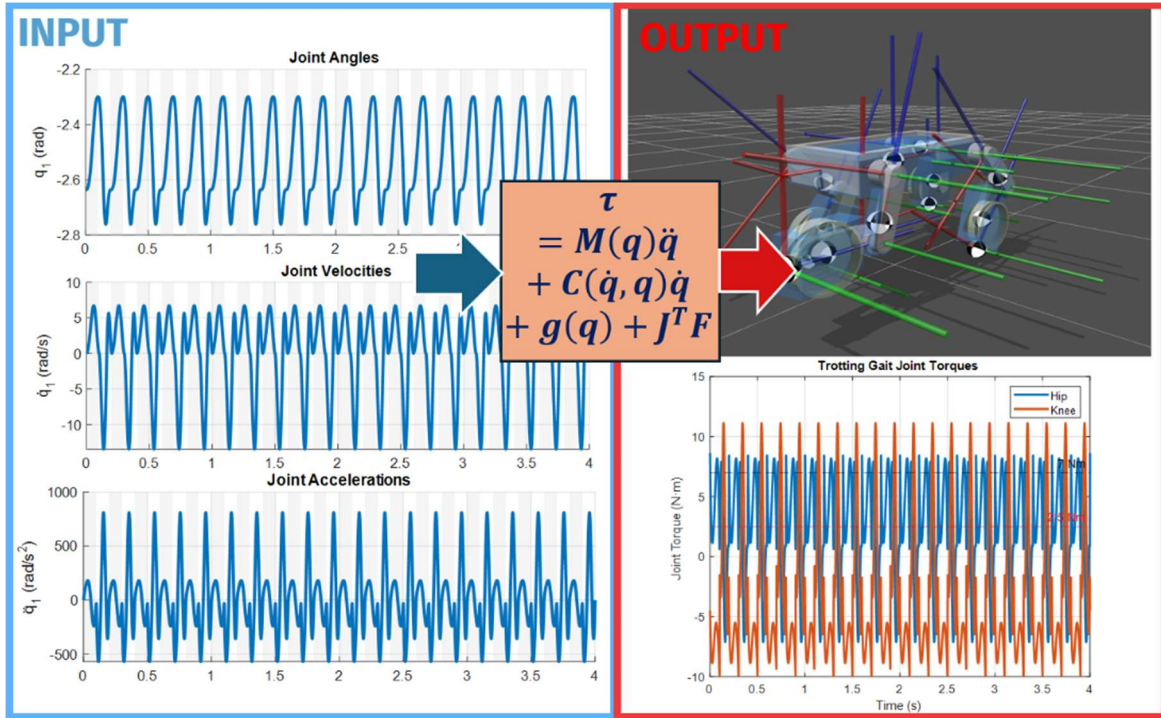
Angular Standing Up Gait: $\theta(t) = \theta_0 + (\theta_f - \theta_0)(10\tau^3 + 6\tau^5 - 15\tau^4)$, $\tau = \frac{t}{T}$

Joint	Initial (rad)	Final (rad)
Hip (θ_1)	-0.50	-1.4
Knee (θ_2)	-2.2	-0.2



Hip: $\theta_1(t) = \theta_{1,mean} + A_1 \sin(\phi)$ $A_1 = 0.4 \text{ rad}, \theta_{1,mean} = -0.8 \text{ rad}$
 Knee: $\theta_2(t) = \theta_{2,mean} + A_2 \sin(\phi + \delta)$ $A_2 = 0.65 \text{ rad}, \theta_{2,mean} = -1.5 \text{ rad}, \delta = 0.4 \text{ rad}$
 $\phi = \frac{2\pi}{T}t, T = 0.2 \text{ seconds}$

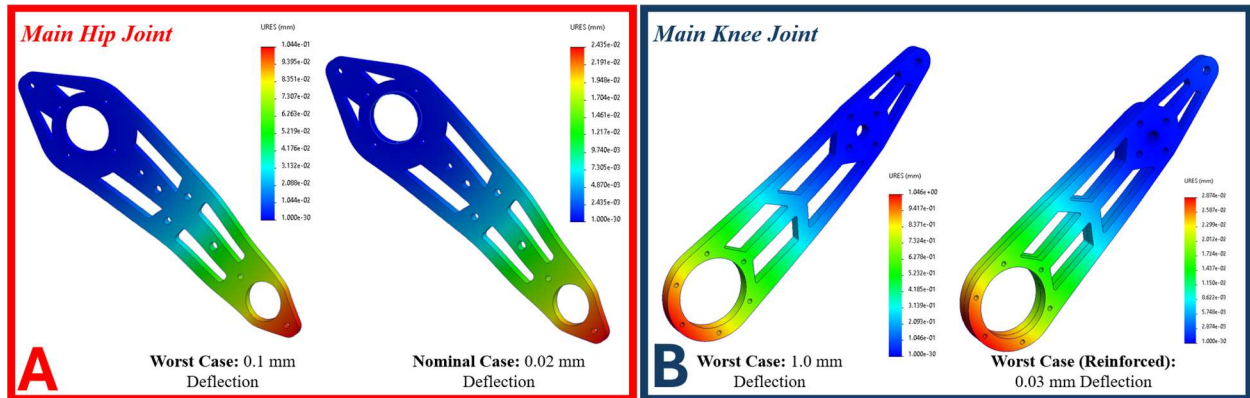
Appendix T: Pinocchio simulation environment allowing for full inverse dynamic simulation to achieve relevant joint torques over walking and standing gaits.



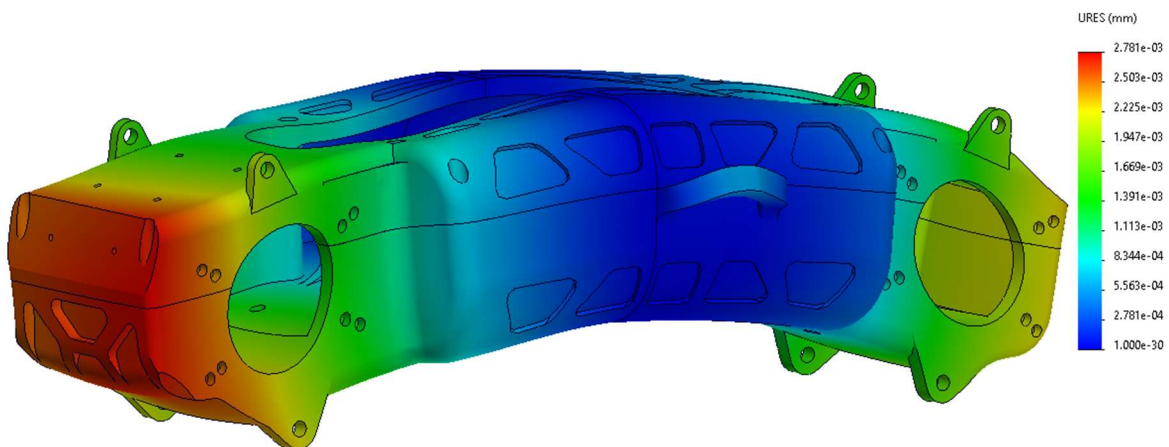
Appendix U: Relevant properties of 6061 T6 Aluminum used for manufacturing and SolidWorks FEA

Properties	AA6061-T6 (aged)
Yield stress (σ_y), MPa	276
Ultimate stress (σ_u), MPa	310
Fatigue limit (σ_f), MPa	96.5
Rupture strain, %	17
Elasticity Modulus (E), GPa	68.9
Poisson's ratio (ν)	0.33
Density (ρ), g/cm ³	2.7

Appendix V: FEA analyses of linkages of interest and main chassis



FEA of linkages of interest and their deflection heatmap. A) Showcases the bearing side of the main hip joint under nominal and max loads. B) Shows the worst case for the knee joint iteration I and iteration II.



FEA of main chassis. Considered PLA infill (with varying densities and Young's modulus) with less than 0.003 mm deflection.

Appendix W: RealSense software architecture

Camera Script	Function
<i>realsense-camera.py</i>	Processes visual data from Intel RealSense camera for 3D object detection.
<i>realsense_calibration.py</i>	Calibrates and saves RealSense camera parameters for optimal 3D object detection.
<i>realsense_depth.py</i>	Estimates depth from RealSense stereo cameras and creates colorized depth map for visualization.
<i>realsense_bbox3d_utils.py</i>	Creates 3D bounding box estimation using depth data.
<i>run_realsense_3d_segmentation.py</i>	Runs object detection and depth mapping scripts.

Appendix X: Software nodes, their primary function, and libraries that are being used or explored

<i>Layer</i>	<i>Node / Module</i>	<i>Primary Function</i>	<i>Supporting Packages & Libraries</i>
<i>Human-Robot Interaction</i>	orchestrator.py	Serves as state machine coordinating high-level robot behavior based on user input and perception outputs	pytransitions
	handle.py	Receives user commands via button input from the Bluetooth handle.	bleak
	audio_comms.py	Converts system messaging into speech and converts voice commands into text to be used within system.	speech_recognition pytsx3
<i>Motion Planning & Perception</i>	motion_controller.py	Executes scripted route following and local obstacle avoidance for both wheeled and legged motion.	scipy
	obstacle_detection.py	Detects obstacles that interfere with traversal capability. (e.g. cannot drive through a scooter, must route around)	PyTorch Torchvision OpenCV
	env_perception.py	Classifies and describes surroundings and off path hazards for user awareness.	Ultralytics YOLOv11 matplotlib pyrealsense2 laser_geometry python-can
<i>Hardware Control & State Estimation</i>	can_motor_driver.py	Translates desired motion commands into CAN messages for low level actuator communication and provides encoder feedback for state estimation.	
	realsense2_camera.py	Interfaces with Intel RealSense d435i to publish image and depth data	librealsense
	rplidar.py	Interfaces with A2M12 RPLIDAR to publish scan data	rplidar
	robot_state_publisher.py	Computes transformations from the world frame to relevant positions within the robot.	

Appendix Y: Anthropomorphic measurements for handle design based off the ANSUR II database

<i>Feature</i>	<i>5th Percentile (Female)</i>	<i>95th Percentile (Male)</i>	<i>Design Measurement</i>	<i>Rationale</i>
<i>Handle Length</i>	70 mm (Palm Length)	120 mm (Palm length)	120 mm	Chosen for full hand engagement of the largest users (95th percentile male).
<i>Handle Diameter</i>	33 mm (Grip)	44 mm (Grip)	35 to 38 mm	Optimized for a "Power Grip" in tools - small enough for the 5th percentile female but also large enough for 95th percentile men

Appendix Z: Societal Impact Assessment

Goal and Scope

<i>Objective of Assessment</i>	<i>Design Function</i>	<i>Functional Unit</i>	<i>Lifecycle Stages Considered</i>	<i>Associated Activities</i>
Assess the social impacts of a robotic guide dog for visually impaired individuals	Provide guidance and navigation to enhance autonomous urban travel for the blind & visually impaired community	1 robotic dog (with handle interface)	Product Use	Daily Guidance, Obstacle Detection, and Maintenance
			Manufacturing	Component Sourcing & Assembly

Inventory Analysis

<i>Product Lifecycle Stage</i>	<i>Stakeholder Group</i>	<i>Social Impact Category</i>	<i>Impact Indicators</i>
Product Use	Consumers	Health and Safety	Number of reported injuries while using device
			Compliance with safety standards and regulations
Manufacturing	Value-chain actors	Supplier Relationships	Supplier diversity (% of small or local suppliers)
			Stability of supplier relationships
	Workers	Hours of Work	Average weekly working hours
			Work satisfaction with work-life balance

Predicted Impact on Human Well-Being

<i>Positive Impact</i>	<i>Negative Impacts</i>
Enhanced mobility and independence for visually impaired individuals	Potential dependency on technology
Increased public awareness of accessibility needs	User frustration if the device malfunctions
BVIP access to certain spaces where guide dogs are prohibited	Social stigmas around robotic assistance devices

Steps to Minimize Negative Social Impacts

To mitigate the negative impacts identified, several measures can be implemented across the manufacturing and product use stages.

During the manufacturing stage, prioritizing ethically sourced materials and selecting suppliers that adhere to international labor and environmental standards helps ensure responsible production. This not

only addresses broader social concerns but also improves product quality and reliability. In turn, this reduces the likelihood of malfunctions that could lead to user frustration.

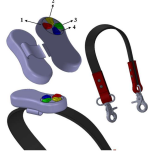
During the product use stage, designing the system with a modular architecture allows for easy repair and component replacement. This directly reduces user frustration by enabling quick maintenance and minimizing downtime if the robot were to malfunction. Additionally, integration of user-friendly features through customizable UI/UX platforms and tactile button controls can reduce overdependence by ensuring the device supports user capability instead of replacing it. To address social stigmas, the device should be designed to be compact and discreet.

Consumers, value-chain actors, and workers were chosen as the stakeholders because they are directly affected across product use and manufacturing stage. Consumers were identified as key stakeholders because they interact directly with the device during the product use stage. Issues such as device malfunction, overdependence, and social stigma primarily affect this group.

Value-chain actors were selected due to their involvement in sourcing materials and producing components during the manufacturing stage. Their practices touch on ethical sourcing, environmental responsibility, and product quality.

Workers were considered stakeholders in this analysis because they are directly impacted by workplace conditions during the manufacturing process. Ensuring ethical production practices helps minimize negative social impacts on this group.

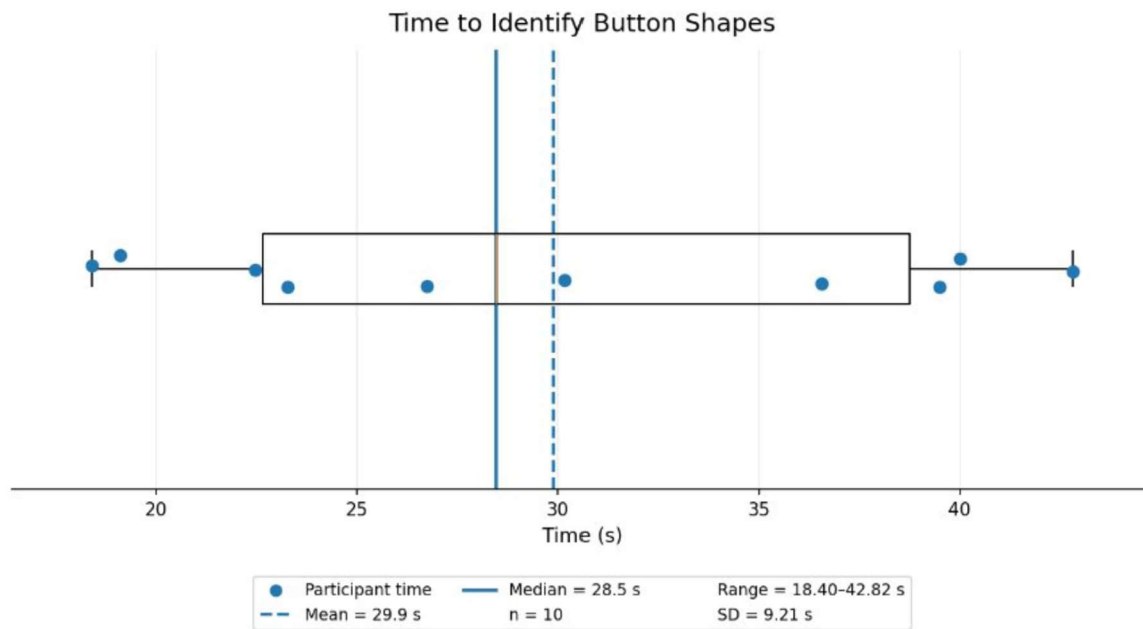
Appendix AA: Handle Concept Design Pros and Cons for Down-Selection

	Concept 1:  Retractable Dog Leash	Concept 2  Rigid Handle	Concept 3  Pull strap Handle with Attachable Remote
Pros	Combined toggle button reduces thumb strain Large central button lowers error rate Flanking buttons minimize thumb travel	Stabilizes center of mass and improves directional feedback Familiar architecture reduces learning curve Single passive DOF enables pick up in same orientation Quick tactile button access Adjustable height via spring-loaded pins to accommodate for	Attaches to most existing handle types Can clip to belt loop independently D-pad configuration minimizes finger travel Familiar remote layout shortens learning curve
Cons	Leash slack delays proprioceptive feedback Hard to reorient quickly if dropped	Tied to one form factor, less adaptable Bulkier than other concepts	Flexible handle attachment reintroduces proprioceptive issues Risk of losing a separate component Less integrated feel than a purpose-built handle

Appendix AB: Feedback from first round user testing

Handle Type	Proprioceptive Feedback “Rate how well you were able to feel when the robot stopped movement” (1 - Cannot discern, 5 - Can easily discern)	Grip Security “Rate how secure your grip feels” (1- Low, 5 - High)	Comfort How comfortable is this to hold? (1- Low, 5 - High)	Button Identification Ability to identify button shape in less than 1 minute by feel alone (Y/N)
Concept 2: Pull Strap Handle	3	4	4	3
Concept 3: Rigid Handle	5	3	3	3

Appendix AC: Time Spent to Identify Button Shapes on Updated Rigid Handle Prototype



Appendix AD: User preference components and framework.

The user interface has two categories:

1. *Communication and Environmental Contextualization*: recreating the feeling of a sighted guide as opposed to autonomous navigation.
2. *Feedback and Control*: allowing the user to acknowledge or give permission before tasks are executed.

The goal of the user interface is to make the user feel more comfortable and safer when navigating with Sidewalk Sidekick.

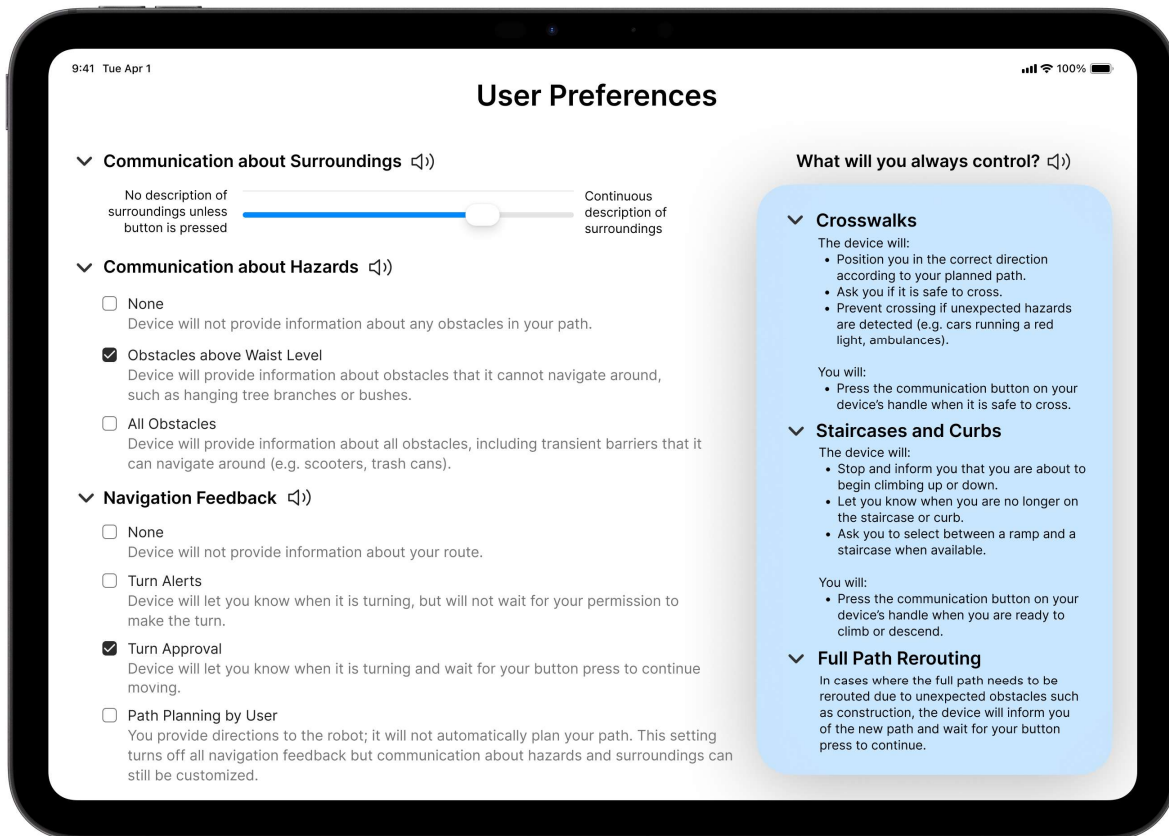
BVIP have varied levels of independence and comfort with assistive technologies besides the white cane; therefore, it was important to ensure that the level of user control and communication from the device was customizable. Therefore, the current prototype of the communication protocol has three categories for user-set preferences as well as tasks that the user will control regardless of what preferences they select.

The figure below is a mock UI of the user control and communication interface. In the final prototype, this would be embedded in a web or mobile application. When the user first uses the device, they will work with an assistive technology specialist or independently enter their preferences. These preferences can then be adjusted as the user gets more comfortable with their regular paths or navigation with the device. The user-set preferences are divided into three main categories:

1. *Communication about Surroundings*: the user will be able to select how much information they want to receive about their surroundings while using the device. The UI is set to a sliding scale to increase intuitiveness, but the level of communication will be discretized in software. Additionally, regardless of what communication preferences the user sets, they will always be able to request an impromptu description of their surroundings using the button on the device's handle.
2. *Communication about Hazards*: users will be able to select whether they would like to receive information about only obstacles above chest level (such as tree branches) or all obstacles, including ones that the device will navigate around (like trash cans or scooters).
3. *Navigation Feedback*: users can ask the device to let them know when it plans to turn or request their permission before turning through a button press. Additionally, more independent users can elect to plan the path themselves. In this situation, the device would primarily focus on obstacle detection and proprioceptive guidance. As it is redundant for the device to let the user know when it is turning in this situation, navigation feedback will be turned off if 'Path Planning by User' is selected. However, all communication settings will still be customizable.

In addition to the customizable settings, the user will always: (1) tell the device when it is safe to cross on a crosswalk, (2) provide acknowledgement before going up or down a staircase or curb, and (3) provide acknowledgement when the full path needs to be rerouted due to unavoidable obstacles like construction. Flowcharts for each of these control workflows are in Appendices N, O, and P.

Primary feasibility of the user control and communication protocol will be evaluated through Experiment 2 in *Feasibility Assessment and Evaluated Criteria*.



Appendix AE: Detailed overview of second-round user testing (Experiment 1)

The relevant variables for Experiment 1 in the context of testing with Raveena are:

- **Independent variables:** number of obstacles (1 or 2), location of obstacles relative to each other (within 2m of each other on opposite lateral sides of the path or on opposite ends of the path), location relative to object's planned path (center of path or right/ left lateral side)
- **User-focused dependent variables:**
 - **A** - How safe do you feel while the robot navigates obstacles? (1 – 5)
 - **B** - Did you feel nervous at any point during the trial? (0 – no, 1 – yes (at what point?))
 - **C** - How smooth did the robot's motion feel? (1 – 5)
 - **D** - Did you ever feel like you wanted to stop or override the robot? (0 – no, 1 – yes (at what point?))
 - **E** - How well were you able to understand the robot's path changes? (1 – 5)

Descriptions of the three trials tested as part of Experiment 1 are detailed in the table below.

<i>Trial</i>	<i>O1</i>	<i>O2</i>	<i>Relative Layout</i>	<i>Relative Location</i>
<i>1</i>	Scooter	N/A	N/A	Center of path
<i>2</i>	Trash bag	N/A	N/A	Left lateral side
<i>3</i>	Scooter	Trash bag	Within 2m of each other, opposite lateral sides	One center, one right

The resulting quantitative feedback from Raveena is detailed in the table below.

<i>Trial</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>1</i>	5	0	4	0	4
<i>2</i>	5	0	4	0	4
<i>3</i>	4	1 (jittery motion in between obstacles)	3	0	3
<i>AVG.</i>	4.6	0.3	3.6	0	3.6

Appendix AF: Estimated runtime for different states.

Values are estimated from RMS torque and motor current curves on the CubeMars specification sheets.

Configuration	$K_T N$ (N m/A)	Current per Actuator (A)	Number of Actuators	Total Current Draw (A)	Estimated Runtime
Walking (AK45-36) only	$0.11 \times 36 = 3.96$	$\frac{6.805}{3.96} = 1.72$	8	$8 \times 1.72 = 13.76$	$\frac{5.0}{13.76} = 0.36 \text{ h} = 21.8 \text{ min}$
Rolling (AK45-10) only	$0.127 \times 10 = 1.27$	$\frac{1.4}{1.27} = 1.10$	4	$4 \times 1.10 = 4.41$	$\frac{5.0}{4.41} = 1.13 \text{ h} = 68.0 \text{ min}$

Appendix AG: MVP design criteria & evaluation results

<i>MVP Design Criteria</i>	<i>Evaluation Results</i>
Detects relevant stationary hazards in front of the user from ground level (0 ft) up to 6ft 6in (2m)	Needs improvement – RealSense FOV too small to cover entire range at closer robot distances. Final design will require downward and upward facing cameras.
Detects hazards up to 3m ahead and as close as 1.6m in front of the guidance device	Meets criteria
Provides guided assistance on a pre-set mock urban route with variable obstacle placement	Meets criteria
Alerts the user of hazards at least 2 seconds before they reach the hazard location	Meets criteria
Device volume $\leq 350 \times 400 \times 350$ mm (L x W x H)	Final prototype dimensions: 300 x 550 x 300 mm This was found to be more reasonable dimensions for the robot, through final iterations could cut volume with more compact integration
Tethering platform and handle design conform to existing interaction strategies	Meets criteria (utilizes traditional rigid handle form factor)
Device weight $< 15\text{kgf}$	Meets criteria (device weight = 14.2 kgf)
Device operates in representative urban scenarios	Meets criteria (demonstrated through rolling and walking gaits).
Device supports motion of at least 1.24 m/s	Meets criteria (maximum speed = 2.42 m/s)
Minimizes cognitive load using different levels for control and feedback during guidance	Meets criteria
Includes proprioceptive and auditory cues during guidance	Meets criteria
Battery life sufficient to support planned experiments	Meets criteria (battery able to last multiple trials and hours of testing)

Appendix AH: Robot-only obstacle avoidance testing results by obstacle size class.

<i>Obstacle Height</i>	<i>Obstacle Width</i>	<i>Avoidance Success</i>
Tall	Thin	80%
Tall	Wide	90%
Short	Thin	60%
Short	Wide	80%

Appendix AI: Detailed overview of experiments

Experiment 1 (Stationary Object Detection and Avoidance): The experimental setup will include several trials, with each trial varying the size (projected area) of the obstacles and their location relative to the robot's planned path. Objects that occupy approximately the same area are treated equivalently by the robot during path planning, regardless of shape. Therefore, varying the shape of the obstacles yields less conclusive results about the robot's guidance ability than varying their size and location. The start and end states of the path will remain constant and the robot will be initially programmed with a straight path. When an obstacle is detected, the robot will autonomously identify and execute the correct path to guide the user around it. Size and location of the obstacles are the independent variables; the dependent variables that will be assessed are obstacle detection success (%), the detection-to-alert time for the user, obstacle avoidance success (%), and user confidence while the robot is rerouting. The controlled variables will include the start and end position of the test and the communication preferences selected by the user.

Experiment 2 (Shared-Control Guidance and Turn Execution): The user starts at a designated position and initiates movement using the "Start route" button on the handle. The robot will move forward and alert the user when the end of the current path segment is reached. It then prompts the user to choose a direction to turn (right or left). After pressing the "Audio command" button on the handle, the user will verbally respond to this prompt and the robot will execute the command. The independent variables that will be changed will be the turn radius and path width. The dependent variables that will be assessed are user confidence and comfort, command and response latency, and route completion success. The control variables are the robot speed, start position, and the pre-set dictionary of voice commands.

Experiment 3 (Traversal Mechanism over Uneven Terrain): The robot will approach an obstacle, detect it and its height, and determine how to traverse the obstacle. The independent variable will be obstacle height and the dependent variables will be the height detection success (%), traversal success (%), and method of traversal.

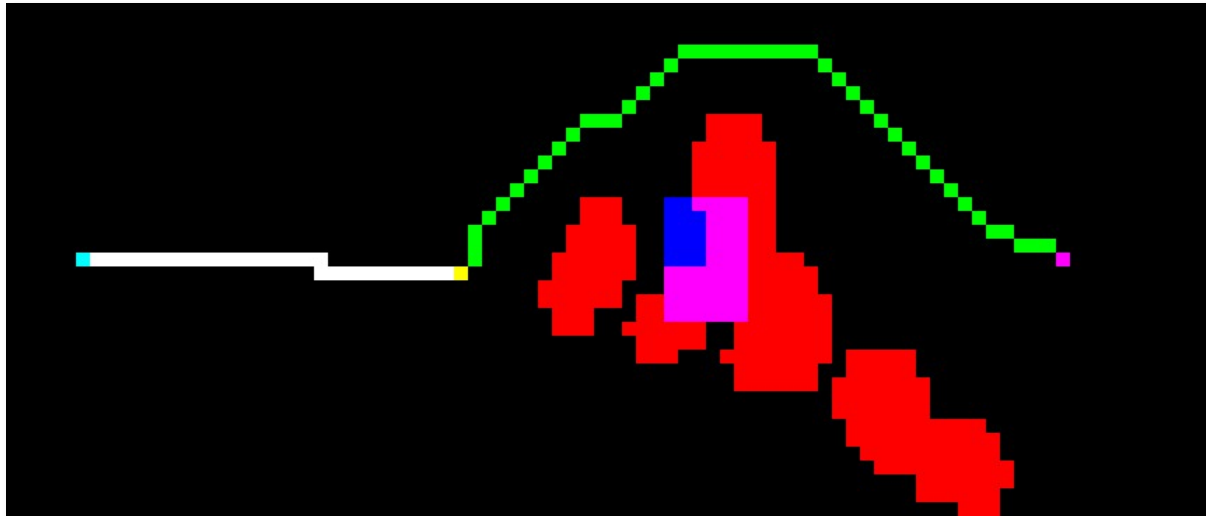
Appendix AJ: Tire design iteration

A) Initial tire design. B) Improved tire design for smoother turning.



Appendix AK: Path planning algorithm iteration

Initially implemented A* algorithm with global map generation (green path is avoidance):



Visualization of local costmap used in final perception stack:

